Exercise-1

Marked Questions can be used as Revision Questions.

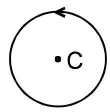
PART - I : SUBJECTIVE QUESTIONS

Section (A) : Magnet and Magnetic field due to a moving charge

A-1. The magnetic moment of a short dipole is, 1 A m^2. What is the magnitude of the magnetic induction in air at 10 cm from centre of the dipole on a line making an angle of 30º from the axis of the dipole?

A-2. A point charge q = 2μC is at the origin. It has velocity $2\hat{i}$ m/s. Find the magnetic field at the following points in vector form (at the moment when the charged particle passes through the origin) :
(i) (2, 0, 0) (ii) (0, 2, 0) (iii) (0, 0, 2) (iv) (2, 1, 2)
(v) Is the magnitude of the magnetic field on the circumference of the circle (in yz plane) $y^2 + z^2 = c^2$ where 'c' is a constant is same every where. Is it same in direction also.
(vi) Answer the above (v) for the circle of same equation but in a plane x = a where 'a' is a constant.

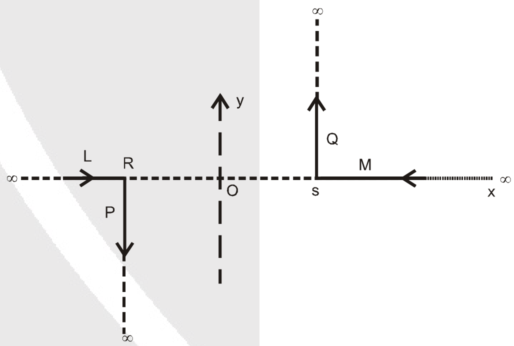
A-3. A particle of negative charge of magnitude 'q' is revolving with constant speed ' V' in a circle of radius 'R' as shown in figure. Find the magnetic field (magnitude and direction) at the following points :



(i) centre of the circle (magnitude and direction)
(ii) a point on the axis and at a distance 'x' from the centre of the ring (magnitude only). Is its direction constant all the time?

Section (B) : Magnetic field due to a straight wire

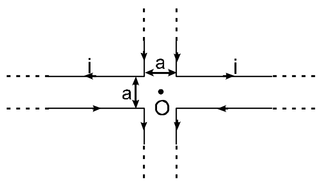
B-1. A pair of stationary and infinitely long bent wires is placed in the X-Y plane as shown in figure. The wires carry currents of 10A each as shown. The segments L and M are along the x-axis. The segments P and Q are parallel to the Y-axis such that OS = OR = 0.02 m. Find the magnitude and direction of the magnetic induction at the origin O.



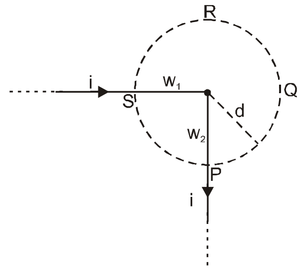
B-2. A current of 1 amp is flowing in the sides of an equilateral triangle of side 4.5 x 10^{-2}m. Find the magnetic field at the centroid of triangle.

B-3. Two straight infinitely long and thin parallel wires are spaced 0.1 m apart and carry a current of 10 ampere each. Find the magnetic field at a point distant 0.1 m from both wires in the two cases when the currents are in the
(i) Same and
(ii) Opposite direction.

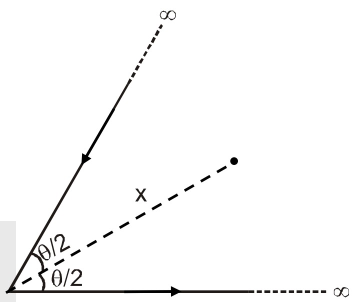
B-4. Four infinitely long 'L' shaped wires, each carrying a current i have been arranged as shown in the figure. Obtain the magnetic field strength at the point 'O' equidistant from all the four corners.



B-5. Figure shows a long wire bent at the middle to form a right angle. Show that the magnitudes of the magnetic fields at the points Q and R are unequal and find these magnitudes. The wire w_1 and the circumference of circle are coplanar and w_2 is perpendicular to plane of paper. Also find the ratio of field at Q and R.

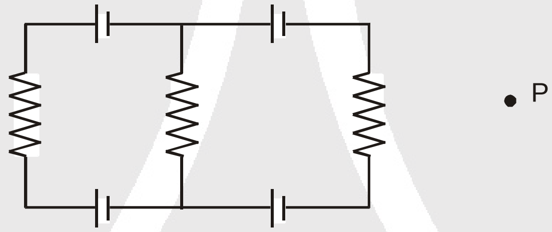


B-6. A long wire carrying a current i is bent to form a plane angle θ. Find the magnetic field B at a point on the bisector of this angle situated at a distance x from the vertex is written in the form of $K\cot\frac{\theta}{4}$ Tesla. Then, find the value of K.



B-7. Find the magnetic field B at the centre of a square loop of side 'a', carrying a current i.

B-8. Each of the batteries shown in figure has an emf equal to 10 V. Find the magnetic field B at the point p.



Section (C) : Magnetic field due to a circular loop

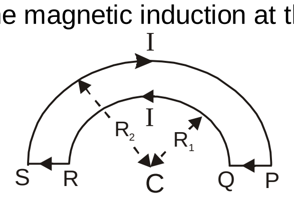
C-1. (i) Two circular coils of radii 5.0 cm and 10 cm carry equal currents of 1 A. The coils have 50 and 100 turns respectively and are placed in such a way that their planes as well as the centre coincide. Find the magnitude of the magnetic field B at the common centre when the currents in the coils are (a) in the same sense (b) in the opposite sense.

(ii) If the outer coil of the above problem is rotated through 90º about a diameter, what would be the magnitude of the magnetic field B at the centre?

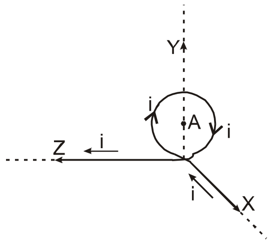
C-2. Two circular coils of wire each having a radius of 4 cm and 10 turns have a common axis and are 6 cm apart. If a current of 1 A passes through each coil in the opposite direction find the magnetic induction.
(a) At the centre of each coil ;
(b) At a point on the axis, midway between them.

Section (D) : Magnetic field due to a straight wire and circular arc

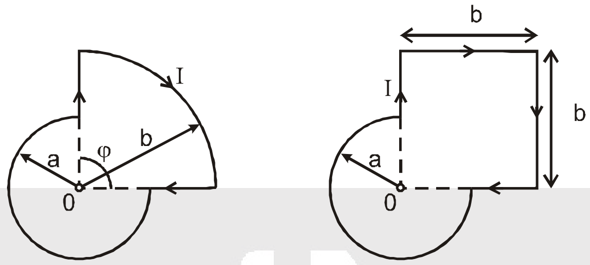
D-1. Two wire loops PQRSP formed by joining two semicircular wires of radii R_1 and R_2 carries a current I as shown in (fig.) The magnitude of the magnetic induction at the center C is



D-2. Find the magnitude of the magnetic induction B of a magnetic field generated by a system of thin conductors (along which a current i is flowing) at a point A (0, R, 0), that is the centre of a circular conductor of radius R. The circular part is in yz plane.



D-3. Find the magnetic induction of the field at the point O of a loop with current I , whose shape is illustrated in figure

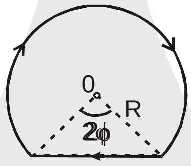


(a) (b)

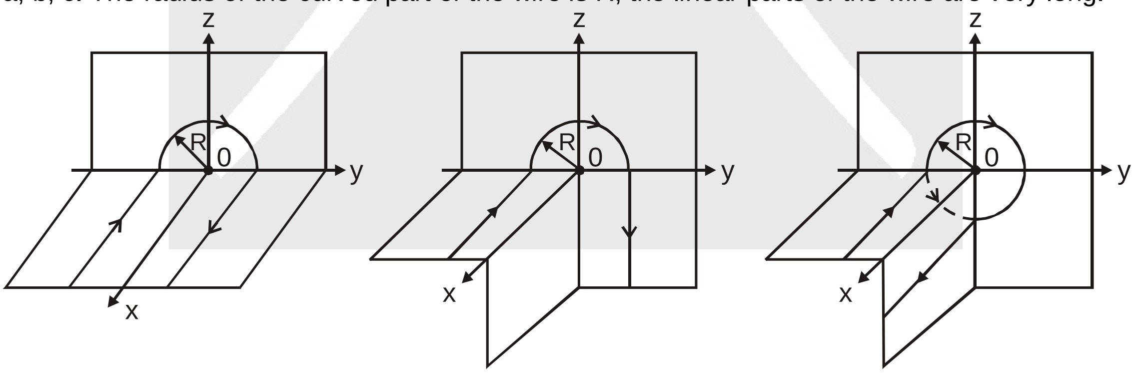
(a) In figure 'a' the radii a and b, as well as the angle φ are known,

(b) In figure b, the radius a and the side b are known.

(c) A current $I = 5.0$ A flows along a thin wire shaped as shown in figure. The radius of a curved part of the wire is equal to R = 120 mm, the angle $2\varphi = 90°$. Find the magnetic induction of the field at the point O.



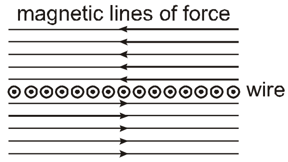
D-4. Find the magnetic induction at the point O if the wire carrying a current I has the shape shown in figure a, b, c. The radius of the curved part of the wire is R, the linear parts of the wire are very long.



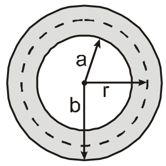
(a) (b) (c)

Section (E) : Magnetic field due to a cylinder, large sheet, solenoid, toroid and ampere's law

E-1. A conductor consists of an infinite number of adjacent wires, each infinitely long & carrying a current i. Show that the lines of B will be as represented in figure & that B for all points in front of the infinite current sheet will be given by, $B = (1/2)\mu_0 ni$, where n is the number of conducting wires per unit length.



E-2. Figure shows a cylindrical conductor of inner radius a & outer radius b which carries a current *i* uniformly spread over its cross section. Show that the magnetic field B for points inside the body of the conductor (i.e. $a < r < b$) is given by, $B = \dfrac{\mu_0 i}{2\pi(b^2 - a^2)} \dfrac{r^2 - a^2}{r}$. Check this formula for the limiting case of $a = 0$.



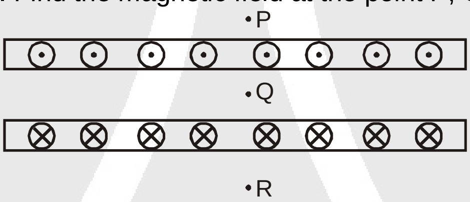
E-3. A thin but long, hollow, cylindrical tube of radius r carries a current i along its length. Find the magnitude of the magnetic field at a distance r/4 from the surface (a) inside the tube (b) outside the tube.

E-4. The magnetic field B inside a long solenoid, carrying a current of 10 A, is 3.14×10^{-2} T. Find the number of turns per unit length of the solenoid.

E-5. A copper wire having resistance 0.01 ohm in each metre is used to wind a 400 turn solenoid of radius 1.0 cm and length 20 cm. Find the emf of a battery which when connected across the solenoid will cause a magnetic field of 1.0×10^{-2} T near the centre of the solenoid.

E-6 A tightly-wound, long solenoid is kept with its axis parallel to a large metal sheet carrying a surface current. The surface current through a width dl of the sheet is Kdl and the number of turns per unit length of the solenoid is n. The magnetic field near the centre of the solenoid is found to be zero. (a) Find the current in the solenoid. (b) If the solenoid is rotated to make its axis 60º to the plane of metal sheet, what would be the magnitude of the magnetic field near its centre?

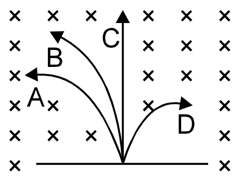
E-7. Two large metal sheets carry surface currents as shown in figure. The current through a strip of width dl is Kdl where K is a constant. Find the magnetic field at the point P, Q and R.



Section (F) : Magnetic force on a charge (Normal incidence)

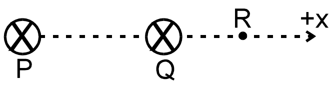
F-1. A charged particle is accelerated through a potential difference of 24 kV and acquires a speed of 2×10^6 m/s. It is then injected perpendicularly into a magnetic field of strength 0.2 T. Find the radius of the circle described by it.

F-2. A neutron, a proton, an electron and an α-particle enters a uniform magnetic field with equal velocities. The field is directed along the inward normal to the plane of the paper. Which of these tracks followed are by electron and α-particle.



F-3. In the formula $X = 3\ YZ^2$, the quantities X and Z have the dimensions of capacitance and magnetic induction respectively. The dimensions of Y in the MKS system are.............

F-4. Two long parallel wires carrying currents 2.5 amps and I amps in the same direction (directed into the plane of the paper) are held at P and Q respectively such that they are perpendicular to the plane of paper. The points P and Q are located at a distance of 5m and 2m respectively from a collinear point R.

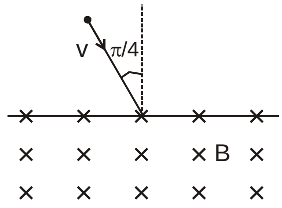


(a) An electron moving with a velocity of 4×10^5 m/s along the positive X-direction experiences a force of magnitude 3.2×10^{-20} N at the point R. Find the value of I.
(b) Find all the positions at which a third long-parallel wire carrying a current of magnitude 2.5 A may be placed so that the magnetic induction at R is zero.

F-5. A magnetic field of $8\hat{k}$ mT exerts a force of $(4.0\hat{i} + 3.0\hat{j}) \times 10^{-10}$ N on a particle having a charge of 5×10^{-10} C and going in the X – Y plane. Find the velocity of the particle.

F-6. An experimenter's diary reads as follows; "a charged particle is projected in a magnetic field of $(7.0\hat{i} - 3.0\hat{j}) \times 10^{-3}$ T. The acceleration of the particle is found to be $(x\hat{i} + 7.0\hat{j}) \times 10^{-6}$ m/s^2. Find the value of x.

F 7. A particle of mass m and positive charge q, moving with a uniform velocity v, enters a magnetic field B as shown in figure. (a) Find the radius of the circular arc it describes in the magnetic field. (b) Find the angle subtended by the arc at the centre. (c) How long does the particle stay inside the magnetic field? (d) Solve the three parts of the above problem if the charge q on the particle is negative.

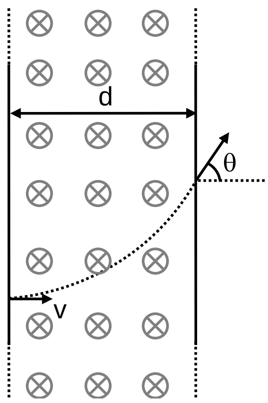


F 8. A particle of mass m and charge q is projected into a region having a perpendicular magnetic field B. Find the angle of deviation (figure) of the particle as it comes out of the magnetic field if the width d of the region is very slightly smaller than

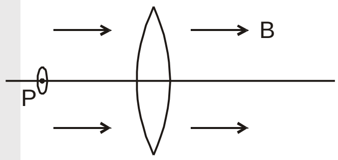
(a) $\dfrac{mv}{qB}$

(b) $\dfrac{mv}{\sqrt{2}qB}$

(c) $\dfrac{3mv}{qB}$



F-9. Figure shows a convex lens of focal length 10 cm lying in a uniform magnetic field B of magnitude 1.2 T parallel to its principal axis. A particle having a charge 2.0×10^{-3} C and mass 2.0×10^{-5} kg is projected perpendicular to the plane of the diagram with a speed of 4.8 m/s. The particle moves along a circle with its centre on the principal axis at a distance of 15 cm from the lens. The axis of the lens and of the circle are same. Show that the image of the particle goes along a circle and find the radius of that circle.



F-10. A uniform magnetic field of magnitude 0.20 T exists in space from east to west. A particle having mass 10^{-5} kg and charge 10^{-5} C is projected from south to north so that it moves with a uniform velocity. Find velocity of projection of the particle? (g = 10 m/s^2)

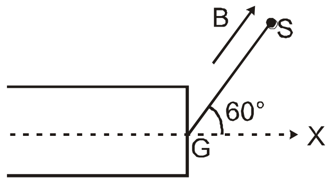
F-11. A tightly-wound, long solenoid has n turns per unit length, a radius r and carries a current i. A particle having charge q and mass m is projected from a point on the axis of the solenoid in a direction perpendicular to the axis. What can be the minimum speed for which the particle should be projected to strike the solenoid?

Section (G) : Magnetic force on a charge (oblique incidence)

G-1. A particle having a charge of 5.0 μC and a mass of 5.0×10^{-12} kg is projected with a speed of 1.0 km/s in a magnetic field of magnitude 5.0 mT. The angle between the magnetic field vector and the velocity vector is $\sin^{-1}(0.90)$. Show that the path of the particle will be a helix. Find the diameter of the helix and its pitch.

G-2. A proton projected in a magnetic field of 0.04 T travels along a helical path of radius 5.0 cm (of projected circle) and pitch 20 cm. Find the components of the velocity of the proton along and perpendicular to the magnetic field. Take the mass of the proton = 1.6×10^{-27} kg.

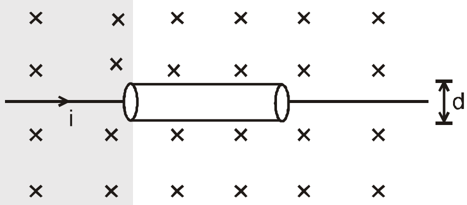
G-3. An electron gun G emits electron of energy 2keV traveling in the positive x-direction. The electrons are required to hit the spot S where GS = 0.1m and the line GS makes an angle of 60^0 with the x-axis, as shown in the fig. A uniform magnetic field $\vec{B}$ parallel to GS exists in the region outside of electron gun. Find the minimum value of B needed to make the electron hit S. [Take mass of electron = 9×10^{-31} kg]



Section (H) : Electric and magnetic force on a charge

H-1. An electron beam passes through a magnetic field of 2×10^{-3} Wb/m^2 and an electric field of 3.2×10^4 V/m, both acting simultaneously. ($\vec{E} \perp \vec{B} \perp \vec{V}$) If the path of electrons remains undeflected calculate the speed of the electron. If the electric field is removed, what will be the radius of the electron path. [mass of electron = 9.1×10^{-31} kg]?

H-2. A current i is passed through a cylindrical gold strip of radius r. The number of free electrons per unit volume is n. (a) Find the drift velocity v of the electrons. (b) If a magnetic field B exists in the region as shown in figure, what is the average magnetic force on the free electrons? (c) Due to the magnetic force, the free electrons get accumulated on one side of the conductor along its length. This produces a transverse electric field in the conductor which opposes the magnetic force on the electrons. Find the magnitude of the electric field which will stop further accumulation of electrons. (d) What will be the potential difference developed across the width of the conductor due to the electron accumulation? The appearance of a transverse emf, when a current-carrying wire is placed in a magnetic field, is called Hall effect.



H-3. A particle moves in a circle of radius 1.0 cm under the action of a magnetic field of 0.40 T. An electric field of 200 V/m makes the path straight. Find the charge/mass ratio of the particle.

H-4. A proton goes undeflected in a crossed electric and magnetic field (the fields are perpendicular to each other) at a speed of 10^5 m/s. The velocity is perpendicular to both the fields. When the electric field is switched off, the proton moves along a circle of radius 2 cm. Find the magnitudes of the electric and the magnetic fields. Take the mass of the proton = 1.6×10^{-27} kg.

H-5. A particle having mass m and charge q is released from the origin in a region in which electric field and magnetic field are given by

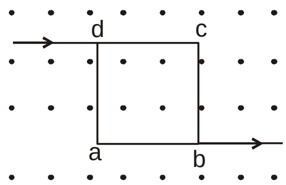
$\vec{B} = +B_0\hat{j}$ and $\vec{E} = +E_0\hat{i}$.

Find the speed of the particle as a function of its X-coordinate.

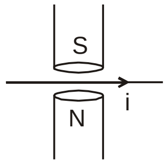
Section (I) : Magnetic force on a current carrying wire

I-1. Consider a 10 cm long portion of a straight wire carrying a current of 10 A placed in a magnetic field of 0.1 T making an angle of 37° with the wire. What magnetic force does the wire experience?

I-2. A current of 2 A enters at the corner d of a square frame abcd of side 10 cm and leaves at the opposite corner b. A magnetic field B = 0.1 T exists in the space in direction perpendicular to the plane of the frame as shown in figure. Find the magnitude and direction of the magnetic forces on the four sides of the frame.



I-3. A magnetic field of strength 1.0 T is produced by a strong electromagnet in a cylindrical region of diameter 4.0 cm as shown in figure. A wire, carrying a current of 2.0 A, is placed perpendicular to and intersecting the axis of the cylindrical region. Find the magnitude of the force acting on the wire.

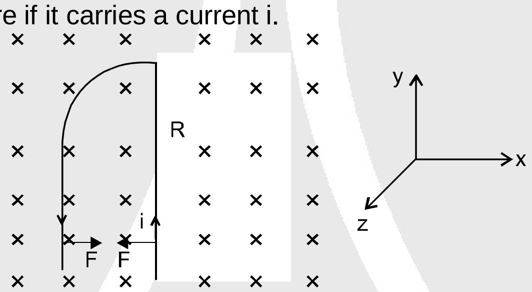


I-4. A wire of length ℓ carries a current i along the y-axis. A magnetic field exists which is given as $\vec{B} = B_0(\hat{i} + \hat{j} + \hat{k})T$. Find the magnitude of the magnetic force acting on the wire.

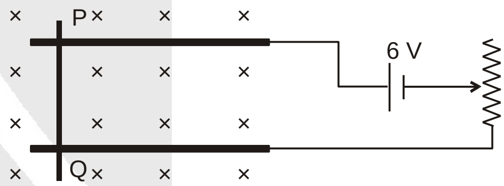
I-5. A thin straight horizontal wire of length 0.2 m whose mass is 10^{-4} kg floats in a magnetic induction field when a current of 10 ampere is passed through it. To make this possible, what should be the minimum magnetic strength? (Take g = 10 m/s^2)

I-6. A wire, carrying a current, i, is kept in the X – Y plane along the curve $y = A \sin\left(\frac{2\pi}{\lambda}x\right)$. A uniform magnetic field B exists in the z-direction. Find the magnitude of the magnetic force on the portion of the wire between x = 0 and x = $\lambda/2$.

I-7. A rigid wire consists of a quarter-circular portion of a circle of radius R and two straight sections (figure). The wire is partially immersed in a perpendicular magnetic field B as shown in the figure. Find the magnetic force on the wire if it carries a current i.



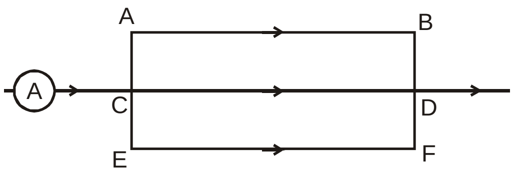
I-8. A metal wire PQ of mass 10 gm lies at rest on two horizontal metal rails separated by 4.90 cm (figure). A vertically downward magnetic field of magnitude 0.800 T exists in the space. The resistance of the circuit is slowly decreased and it is found that when the resistance goes below 20.0 Ω, the wire PQ starts sliding on the rails. Find the coefficient of friction. Neglect magnetic force acting on wire PQ due to metal rails (g = 9.8 m/s^2)



I-9. Two long wires carrying current in the same direction attract each other but two parallel beams of electrons moving in the same direction repel each other. Explain

I-10. The magnetic field existing in a region is given by $\vec{B} = B_0\left(1 - \frac{x}{\ell}\right)\hat{k}$, where B_0 and ℓ are constants, x is the x coordinate of a point and $\hat{k}$ is the unit vector along Z axis. A square loop of edge ℓ and carrying a current i, is placed with its edges parallel to the x, y axes. Find the magnitude of the net magnetic force experienced by the loop.

I-11. Figure shows a part of an electric circuit. The wires AB, CD and EF are very long and have identical resistances. The separation between the neighboring wires is 2 cm. The wires AE and BF have negligible resistance and the ammeter reads 60 A. Calculate the magnetic force per unit length on AB and CD.



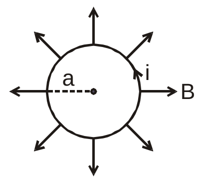
I-12. Two long straight parallel conductors are separated by a distance of r_1 = 5cm and carry constant currents i_1 = 10 A & i_2 = 20 A. What work per unit length of a conductor must be done to increase the separation between the conductors to r_2 = 10 cm if, currents flow in the same direction?

I-13. A straight rod of mass m and length ℓ can slide on two parallel plastic rails kept in a horizontal plane with a separation d. The coefficient of friction between the wire and the rails is μ. If the wire carries a current i, what minimum magnetic field should exist in the space in order to slide the wire on the rails.

Section (J) : Magnetic force and torque on a current carrying loop and magnetic dipole moment

J-1. A circular coil of 100 turns has an effective radius 0.05 m and carries a current of 0.1 amp (assume current remains constant). How much work is required to turn it in an external magnetic field of 1.5 wb/m^2 through 180° about an axis perpendicular to the magnetic field. The plane of the coil is initially perpendicular to the magnetic field.

J-2. (a) A circular loop of radius a, carrying a current i, is placed in a two-dimensional magnetic field. The centre of the loop coincides with the centre of the field (figure). The strength of the magnetic field at the periphery of the loop is B. Find the magnetic force on the wire.



(b) A hypothetical magnetic field existing in a region is given by $\vec{B} = B_0\vec{e}_r$, where $\vec{e}_r$ denotes the unit vector along the radial direction of a point relative to the origin and B_0 = constant. A circular loop of radius a, carrying a current i, is placed with its plane parallel to the X-Y plane and the centre at (0, 0, a). Find the magnitude of the magnetic force acting on the loop.

J-3 A rectangular coil of 100 turns has length 4 cm and width 5 cm. It is placed with its plane parallel to a uniform magnetic field and a current of 2A is sent through the coil. Find the magnitude of the magnetic field B, if the torque acting on the coil is 0.2 N-m.

J-4 A 50-turn circular coil of radius 4 cm carrying a current of 2.5 A is rotated in a magnetic field of strength 0.20 T. (a) What is the maximum torque that acts on the coil? (b) In a particular position of the coil, the torque acting on it is half of this maximum. What is the angle between the magnetic field and the plane of the coil?

J-5 A square loop of sides 10 cm carries a current of 10 A. A uniform magnetic field of magnitude 0.20 T exists parallel to one of the side of the loop. (a) What is the force acting on the loop? (b) What is the torque acting on the loop?

J-6. A circular coil of diameter 2.0 cm has 500 turns in it and carries a current of 1.0 A. Its axis makes an angle of 30º with the uniform magnetic field of magnitude 0.40 T that exists in the space. Find the torque acting on the coil.

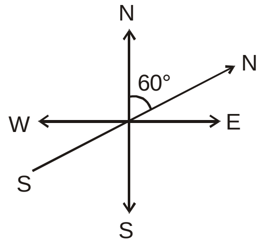
J-7. A circular loop carrying a current i has wire of total length L. A uniform magnetic field B exists parallel to the plane of the loop. (a) Find the torque on the loop. (b) If the same length of the wire is used to form a rectangular loop of side ratio 1 : 2, what would be the torque? Which is larger?

J-8. A charge Q is spread uniformly over an insulated ring of radius R. What is the magnetic moment of the ring if it is rotated with an angular velocity ω about its axis?

Section (K) : Magnetic field due to a magnet and earth

K-1. Two circular coils each of 100 turns are held such that one lies in the vertical plane and the other in the horizontal plane with their centres coinciding. The radius of the vertical and the horizontal coils are respectively 20 cm and 30 cm. If the directions of the current in them are such that the earth's magnetic field at the centre of the coil is exactly neutralized, calculate the current in each coil. [horizontal component of the earth's field = 27.8 μ_0A m^{-1} ; angle of dip = 30º]

K-2. A short magnet of magnetic moment 6 Amp.m^2 is lying in a horizontal plane with its North pole pointing 60º East of North. Find the net horizontal magnetic field at a point on the axis of the magnet 0.2 m away from it. [Horizontal component of earth's magnetic field = 0.3 x 10^{-4} tesla]



K-3. A coil of 50 turns and 20 cm diameter is made with a wire of 0.2 mm diameter and resistivity $2\times10^{-6}\Omega$ cm. The coil is connected to a source of EMF. 20 V and negligible internal resistance.
(a) Find the current through the coil .
(b) What must be the minimum potential difference across the coil so as to nullify the earth's horizontal magnetic induction of 3.14×10^{-5} tesla at the centre of the coil. How should the coil be placed to achieve the above result.

Section (L) : Magnetic material

L-1. A magnetising field of 1600 Am^{-1} produces a magnetic flux of 2.4×10^{-5} wb in an iron bar of cross-sectional area 0.2cm^2. Calculate permeability and susceptibility of the bar.

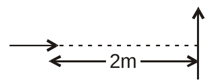
L-2. The core of toroid of 3000 turns has inner and outer radii of 11 cm and 12 cm respectively. A current of 0.6 A produces a magnetic field of 2.5 T in the core. Compute relative permeability of the core. ($\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}).

PART - II : ONLY ONE OPTION CORRECT TYPE

*** Marked Questions are More than One Correct**

Section (A) : Magnet and Magnetic field due to a moving charge

A-1. Two identical short magnetic dipoles of magnetic moments 1.0 A-m^2 each, placed at a separation of 2 m with their axes perpendicular to each other. The resultant magnetic field at a point midway between the dipole is:



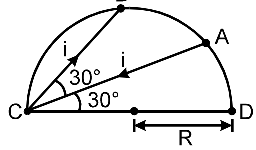
(A) 5×10^{-7} T (B) $\sqrt{5} \times 10^{-7}$ T (C) 10^{-7} T (D) 2×10^{-7} T

A-2. A point charge is moving in a circle with constant speed. Consider the magnetic field produced by the charge at a fixed point P (not centre of the circle) on the axis of the circle.
(A) it is constant in magnitude only (B) it is constant in direction only
(C) it is constant in direction and magnitude both (D) it is not constant in magnitude and direction both.

Section (B) : Magnetic field due to a straight wire

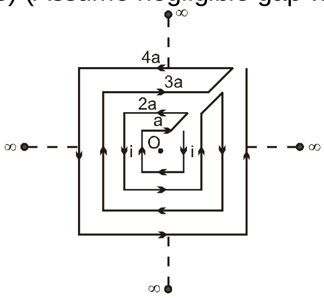
B-1. Two infinitely long, thin, insulated, straight wires lie in the x-y plane along the x and y-axis respectively. Each wire carries a current i, respectively in the positive x-direction and positive y-direction. The magnetic field will be zero at all points on the straight line:
(A) y = x (B) y = – x (C) y = x – 1 (D) y = – x + 1

B-2. A current carrying wire is placed in the grooves of an insulating semi circular disc of radius 'R', as shown. The current enters at point A and leaves from point B. Determine the magnetic field at point D.



(A) $\frac{\mu_0 I}{8\pi R\sqrt{3}}$ (B) $\frac{\mu_0 I}{4\pi R\sqrt{3}}$ (C) $\frac{\sqrt{3}\mu_0 I}{4\pi R}$ (D) none of these

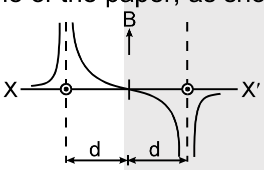
B-3. Determine the magnitude of magnetic field at the centre of the current carrying wire arrangement shown in the figure. The arrangement extends to infinity. (The wires joining the successive squares are along the line passing through the centre) (Assume negligible gap while wire is turned into squares)

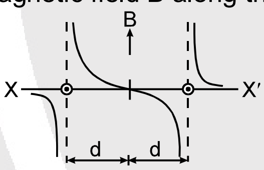


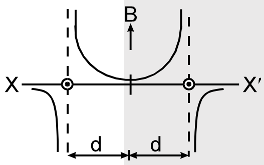
(A) $\frac{\mu_0 i}{\sqrt{2}\pi a}$ (B) 0 (C) $\frac{2\sqrt{2}\mu_0 i}{\pi a}\ell n2$ (D) none of these

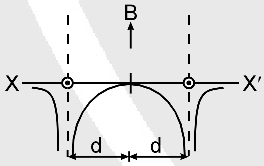
B-4. Two parallel, long wires carry currents i_1 and i_2 with $i_1 > i_2$. When the current are in the same direction, the magnetic field at a point midway between the wire is 20μT. If the direction of i_1 is reversed, the field becomes 30μT. The ratio i_1/i_2 is
(A) 4 (B) 3 (C) 5 (D) 1

B-5. Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper, as shown. The variation of the magnetic field B along the XX' is given by

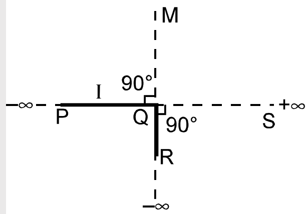
(A) 

(B) 

(C) 

(D) 

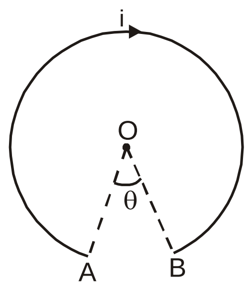
B-6. An infinitely long conductor PQR is bent to form a right angle as shown. A current I flows through PQR. The magnetic field due to this current at the point M is H_1. Now, another infinitely long straight conductor QS is connected at Q so that the current in PQ remaining unchanged. The magnetic field at M is now H_2. The ratio H_1/H_2 is given by



(A) 1/2 (B) 1 (C) 2/3 (D) 2

Section (C) : Magnetic field due to a circular loop, a straight wire and circular arc, cylinder, large sheet, solenoid, toroid and ampere's law

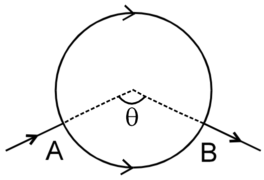
C-1. A current carrying wire AB of the length $2\pi R$ is turned along a circle, as shown in figure. The magnetic field at the centre O.



(A) $\frac{\mu_0 i}{2R}\left(\frac{2\pi-\theta}{2\pi}\right)^2$ (B) $\frac{\mu_0 i}{2R}\left(\frac{2\pi-\theta}{2\pi}\right)$

(C) $\frac{\mu_0 i}{2R}(2\pi-\theta)$ (D) $\frac{\mu_0 i}{2R}(2\pi+\theta)^2$

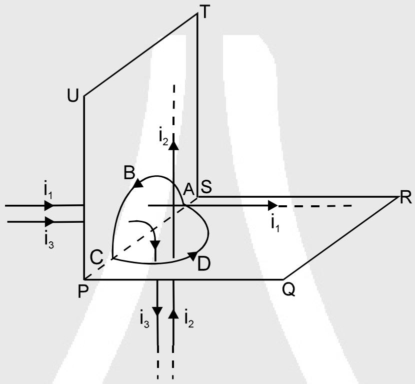
C-2. A battery is connected between two points A and B the circumference of a uniform conducting ring of radius r and resistance R. One of the arcs AB of the ring subtends an angle θ at the centre. The value of the magnetic induction at the centre due to the current in the ring is:



(A) zero, only if $\theta = 180^o$ (B) zero for all values of θ
(C) proportional to $2(180^o - \theta)$ (D) inversely proportional to r

C-3. A wire is wound on a long rod of material of relative permeability $\mu_r = 4000$ to make a solenoid. If the current through the wire is 5 A and number of turns per unit length is 1000 per metre, then the magnetic field inside the solenoid is :
(A) 25.12 mT (B) 12.56 m T (C) 12.56 T (D) 25.12 T

C-4. Figure shows an amperian path ABCDA. Part ABC is in vertical plane PSTU while part CDA is in horizontal plane PQRS. Direction of circumlation along the path is shown by an arrow near point B and at D. $\oint \vec{B}.d\vec{\ell}$ for this path according to Ampere's law will be :



(A) $(i_1 - i_2 + i_3)\,\mu_0$ (B) $(-i_1 + i_2)\,\mu_0$ (C) $i_3\,\mu_0$ (D) $(i_1 + i_2)\,\mu_0$

C-5. A circular loop of radius r carries a current i. How should a long, straight wire carrying a current 3i be placed in the plane of the circle so that the magnetic field at the centre of the loop becomes zero?
(A) At distance $\frac{3r}{\pi}$ from the centre of loop (B) At distance $\frac{2r}{\pi}$ from the centre of loop
(C) At distance $\frac{2r}{3\pi}$ from the centre of loop (D) Not possible

C-6. A long solenoid is fabricated by closely winding a wire of diameter 0.5 mm over a cylindrical nonmagnetic frame so that the successive turns nearly touch each other. What would be the magnetic field B at the centre of the solenoid if it carries a current of 2.5 A ?
(A) $\pi \times 10^{-3}$ T (B) $2\pi \times 10^{-3}$ T (C) $4\pi \times 10^{-3}$ T (D) None of these

C-7. A uniform current I flows along the length of an infinitely long, straight, thin walled pipe. Then
(A) the magnetic field at all points inside the pipe is the same, but not zero
(B) the magnetic field at any point inside the pipe is zero
(C) the magnetic field is zero only on the axis of the pipe
(D) the magnetic field is different at different points inside the pipe.

C-8. A circular loop is kept in that vertical plane which contains the north-south direction. It carries a current that is towards south at the topmost point. Let A be a point on axis of the circle to the east of it and B a point on this axis to the west of it. The magnetic field due to the loop
(A) is towards east at A and towards west at B (B) is towards west at A and towards east at B
(C) is towards east at both A and B (D) is towards west at both A and B

C-9. A long, thick straight conductor of radius R carries current I uniformly distributed in its cross section area. The ratio of energy density of the magnetic field at distance R/2 from surface inside the conductor and outside the conductor is:
(A) 1: 16 (B) 1: 1 (C) 1: 4 (D) 9/16

Section (D) : Magnetic force on a charge (Normal incidence)

D-1. Which of the following particles will experience minimum magnetic force (magnitude) when projected with the same velocity perpendicular to a magnetic field?
(A) Be +++ (B) proton (C) α -particle (D) Li^{++}

D-2. Electric current i enters and leaves a square loop made of homogeneous wire of uniform cross-section through diagonally opposite corners. A charge particle q moving along the axis of the square loop. Passes through centre at speed υ. The magnetic force acting on the particle when it passes through the centre has a magnitude
(A) $qv\frac{\mu_0 i}{2a}$ (B) $qv\frac{\mu_0 i}{2\pi a}$ (C) $qv\frac{\mu_0 i}{a}$ (D) zero

D-3. Two particles X and Y having equal charges, after being accelerated through the same potential difference, enter a region of uniform magnetic field and describe circular paths of radii R_1 and R_2 respectively. The ratio of the masses of X to that of Y.
(A) $\left(\frac{R_1}{R_2}\right)^{1/2}$ (B) $\frac{R_2}{R_1}$ (C) $\left(\frac{R_1}{R_2}\right)^2$ (D) $\frac{R_1}{R_2}$

D-4. A negative charged particle falling freely under gravity enters a region having uniform horizontal magnetic field pointing towards north. The particle will be deflected towards
(A) East (B) West (C) North (D) South

D-5. A proton is moved along a magnetic field line. The magnetic force on the particle is
(A) along its velocity (B) opposite to its velocity
(C) perpendicular to its velocity (D) zero.

D-6. A proton beam is going from west to east and an electron beam is going from east to west. Neglecting the earth's magnetic field, the electron beam will be deflected
(A) towards the proton beam (B) away from the proton beam
(C) away from the electron beam (D) None of these

Section (E) : Magnetic force on a charge (oblique incidence)

E- 1. A proton of mass m and charge q enters a magnetic field B with a velocity v at an angle θ with the direction of B. The radius of curvature of the resulting path is
(A) $\frac{mv}{qB}$ (B) $\frac{mv\sin\theta}{qB}$ (C) $\frac{mv}{qB\sin\theta}$ (D) $\frac{mv\cos\theta}{qB}$

Section (F) : Electric and magnetic force on a charge

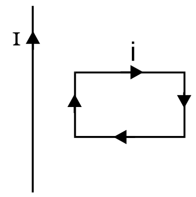
F-1. A positively charged particle moves in a region having a uniform magnetic field and uniform electric field in same direction. At some instant, the velocity of the particle is perpendicular to the field direction. The path of the particle will be
(A) a straight line (B) a circle
(C) a helix with uniform pitch (D) a helix with increasing pitch.

F-2. If a charged particle projected in a gravity-free room it does not deflect,
(A) electric field and magnetic field must be zero
(B) both electric field and magnetic field may be present
(C) electric field will be zero and magnetic field may be zero
(D) electric field may be zero and magnetic field must be zero

Section (G) : Magnetic force on a current carrying wire

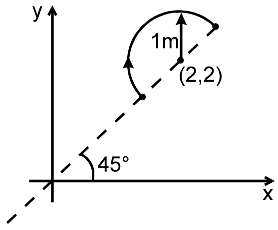
G-1. A conducting circular loop of radius r carries a constant current i. It is placed in a uniform magnetic field **B** such that **B** is perpendicular to the plane of the loop. The magnetic force acting on the loop is
(A) i r B (B) $2\pi r i B$ (C) zero (D) $\pi r i B$

G-2. A rectangular loop carrying a current i is situated near a long straight wire such that the wire is parallel to one of the sides of the loop and the plane of the loop is same of the left wire. If a steady current I is established in the wire as shown in the (fig) the loop will -



(A) Rotate about an axis parallel to the wire
(B) Move away from the wire
(C) Move towards the wire
(D) Remain stationary.

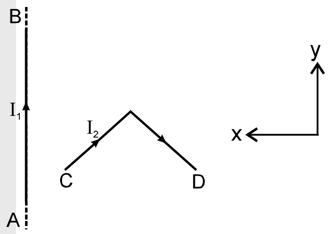
G-3. A uniform magnetic field $\vec{B} = \left(3\hat{i} + 4\hat{j} + \hat{k}\right)$ exists in region of space. A semicircular wire of radius 1 m carrying current 1 A having its centre at (2, 2, 0) is placed in x-y plane as shown in fig. The force on semicircular wire will be



(A) $\sqrt{2}(\hat{i} + \hat{j} + \hat{k})$ (B) $\sqrt{2}(\hat{i} - \hat{j} + \hat{k})$ (C) $\sqrt{2}(\hat{i} + \hat{j} - \hat{k})$ (D) $\sqrt{2}(-\hat{i} + \hat{j} + \hat{k})$

G-4. Select the correct alternative(s):
Two thin long parallel wires separated by a distance 'b' are carrying a current 'i' ampere each. The magnitude of the force per unit length exerted by one wire on the other is
(A) $\frac{\mu_0 i^2}{b^2}$ (B) $\frac{\mu_0 i^2}{2\pi b}$ (C) $\frac{\mu_0 i}{2\pi b}$ (D) $\frac{\mu_0 i}{2\pi b^2}$

G-5. In the figure shown a current I_1 is established in the long straight wire AB. Another wire CD carrying current I_2 is placed in the plane of the paper. The line joining the ends of this wire is perpendicular to the wire AB. The resultant force on the wire CD is:



(A) zero (B) towards negative x-axis
(C) towards positive y-axis (D) none of these

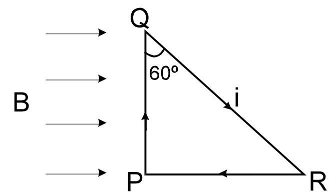
G-6. A steady current 'I' flows in a small square loop of wire of side L in a horizontal plane. The loop is now folded about its middle such that half of it lies in a vertical plane. Let $\vec{\mu}_1$ and $\vec{\mu}_2$ respectively denote the magnetic moments of the current loop before and after folding. Then :

(A) $\vec{\mu}_2 = 0$ (B) $\vec{\mu}_1$ and $\vec{\mu}_2$ are in the same direction
(C) $\frac{|\vec{\mu}_1|}{|\vec{\mu}_2|} = \sqrt{2}$ (D) $\frac{|\vec{\mu}_1|}{|\vec{\mu}_2|} = \frac{1}{\sqrt{2}}$

G-7. A current-carrying, straight wire is kept along the axis of a square loop carrying a current. The straight wire
(A) will exert an inward force on the square loop
(B) will exert an outward force on the squareloop
(C) will not exert any force on the square loop
(D) will exert a force on the square loop parallel to itself.

G-8. Two parallel wires carry currents of 10 A and 40 A in opposite directions. Another wire carrying a current antiparallel to 10 A is placed midway between the two wires. The magnetic force on it will be
(A) towards 20 A (B) towards 40 A
(C) zero (D) perpendicular to the plane of the currents

G-9. For the circuit shown in figure, the direction and magnitude of the force on PQR is :



(A) No resultant force act on the loop (B) ILB out of the page
(C) $\frac{1}{2}$ ILB into the page (D) ILB into the page

SECTION (H) : MAGNETIC FORCE AND TORQUE ON A CURRENT CARRYING LOOP AND MAGNETIC DIPOLE MOMENT

H-1. A bar magnet has a magnetic moment 2.5 JT^{-1} and is placed in a magnetic field of 0.2 T. Work done in turning the magnet from parallel to antiparallel position relative to the field direction is :
(A) 0.5 J (B) 1 J (C) 2.0 J (D) Zero

H-2. A circular loop of area 1 cm^2, carrying a current of 10 A, is placed in a magnetic field of 0.1 T perpendicular to the plane of the loop. The torque on the loop due to the magnetic field is
(A) zero (B) 10^{-4} N-m (C) 10^{-2} N-m (D) 1 N-m

H-3. A rod of length ℓ having uniformly distributed charge Q is rotated about one end with constant frequency ' f '. Its magnetic moment.
(A) $\pi f Q \ell^2$ (B) $\frac{\pi f Q \ell^2}{3}$ (C) $\frac{2\pi f Q \ell^2}{3}$ (D) $2\pi f Q \ell^2$

Section (I) : Magnetic field due to earth

I-1. A power line lies along the east-west direction and carries a current of 10 ampere. The force per metre due to the earth's magnetic field of 10^{-4} T is
(A) 10^{-5} N (B) 10^{-4} N (C) 10^{-3} N (D) 10^{-2} N

I-2. A circular coil of radius 20 cm and 20 turns of wire is mounted vertically with its plane in magnetic meridian. A small magnetic needle (free to rotate about vertical axis) is placed at the center of the coil. It is deflected through 45° when a current is passed through the coil and in equilibrium (Horizontal component of earth's field is 0.34×10^{-4} T). The current in coil is:
(A) $\frac{17}{10\pi}$ A (B) 6A (C) 6×10^{-3} A (D) $\frac{3}{50}$ A

Section (J) : Magnetic material

J-1 The magnetic materials having negative magnetic susceptibility are:
(A) Non magnetic (B) Para magnetic (C) Diamagnetic (D) Ferromagnetic

J-2 When a small magnetising field H is applied to a magnetic material, the intensity of magnetisation (I) is proportional to :

(A) H^{-2} (B) $H^{1/2}$ (C) H (D) H^2

J-3 How does the magnetic susceptibility χ of a paramagnetic material change with absolute temperatue T ?

(A) $\chi \propto T$ (B) $\chi \propto T^{-1}$ (C) χ = constant (D) $\chi \propto e^T$

J-4 Consider the following statements for a paramagnetic substance kept in a magnetic field :

(a) If the magnetic field increases, the magnetisation increases.

(b) If temperature rises, the magnetisation increases.

(A) Both (a) and (b) are true (B) (a) is true but (b) is false

(C) (b) is true but (a) is false (D) Both (a) and (b) are false

J-5. Which of the following relations is not correct ?

(A) $B = \mu_0 (H + I)$ (B) $B = \mu_0 H (1 + \chi_m)$ (C) $\mu_0 = \mu (1 + \chi_m)$ (D) $\mu_r = 1 + \chi_m$

J-6 The hysteresis loop for the material of a permanent magnet is :

(A) short and wide (B) tall and narrow (C) tall and wide (D) short and narrow

J-7 Select the incorrect alternative (s) :

When a ferromagnetic material goes through a complete cycle of magnetisation, the magnetic susceptibility :

(A) has a fixed value (B) may be zero (C) may be infinite (D) may be nagative

J-8 The material for making permanent magnets should have :

(A) high retentivity, high coercivity (B) high retentivity, low coercivity

(C) low retentivity, high coercivity (D) low retentivity, low coercivity

J-9 (a) Soft iron is a conductor of electricity. (b) It is a magnetic material.

(c) It is an alloy of iron.

(d) It is used for making permanent magents. State whether :

(A) a and c are true (B) a and b are true (C) c and d are true (D) b and d are true

J-10 Soft iron is used in many electrical machines for :

(A) low hysteresis loss and low permeability (B) low hysteresis loss and high permeability

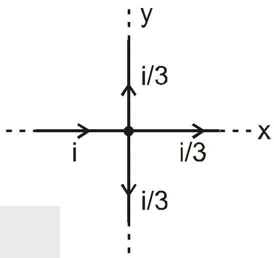
(C) high hysteresis loss and low permeability (D) high hysteresis loss and high permeability

PART - III : MATCH THE COLUMN

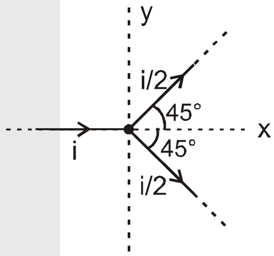
1. Column-II gives four situations in which three (in q, r, s, t) and four (in p) semi infinite current carrying wires are placed in xy-plane as shown. The magnitude and direction of current is shown in each figure. Column-I gives statements regarding the x, y and z components of magnetic field at a point P whose coordinates are P (0, 0, d). Match the statements in column-I with the corresponding figures in column-II and indicate your answer by darkening appropriate bubbles in the 4 × 4 matrix given in OMR.

Column I | **Column II**

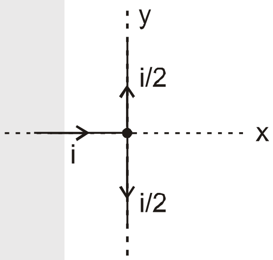
(A) The x component of magnetic field at point P is zero in

(p) 

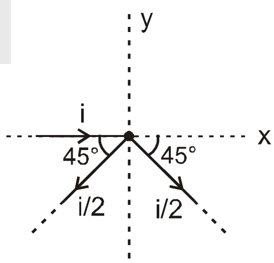
(B) The z component of magnetic field at point P is zero in

(q) 

(C) The magnitude of magnetic field at point P is $\frac{\mu_0 i}{4\pi d}$ in

(r) 

(D) The magnitude of magnetic field at point P is less than $\frac{\mu_0 i}{2\pi d}$ in

(s) 

2. There are four situations given in column I involving a magnetic dipole of dipole moment $\vec{\mu}$ placed in uniform external magnetic field $\vec{B}$. Column II gives corresponding results. Match the situtations in column I with the corresponding results in column II

Column - I	Column - II
(A) Magnetic dipole moment $\vec{\mu}$ is parallel to uniform external magnetic field $\vec{B}$ (angle between both vectors is zero)	(p) force on dipole is zero
(B) Magnetic dipole moment $\vec{\mu}$, is perpendicular to uniform external magnetic field $\vec{B}$	(q) torque on dipole is zero
(C) Angle between magnetic dipole moment $\vec{\mu}$ and uniform external magnetic field $\vec{B}$ is acute	(r) magnitude of torque is (μB)
(D) Angle between magnetic dipole moment $\vec{\mu}$ and uniform external magnetic field $\vec{B}$ is 180°.	(s) potential energy of dipole due to external magnetic field is (μB)
	(t) Magnetic dipole is in stable equilibrium

3. A particle enters a space where exists uniform magnetic field $\vec{B} = B_x\vec{i} + B_y\vec{j} + B_z\vec{k}$ & uniform electric field $\vec{E} = E_x\vec{i} + E_y\vec{j} + E_z\vec{k}$ with initial velocity $\vec{u} = u_x\vec{i} + u_y\vec{j} + u_z\vec{k}$. Depending on the values of various components the particle selects a path. Match the entries of column A with the entries of column B. The components other than specified in column A in each entry are non-zero. Neglect gravity.

COLUMN A	COLUMN B
(A) $B_y = B_z = E_x = E_z = 0$, $u = 0$	(p) circle
	(q) helix with uniform pitch and constant radius
(B) $E = 0$; $u_xB_x + u_yB_y \neq -u_zB_z$	(r) cycloid
(C) $\vec{u} \times \vec{B} = 0, \vec{u} \times \vec{E} = 0$	(s) helix with uniform pitch and variable radius
(D) $\vec{u} \perp \vec{B}, \vec{B} \| \vec{E}$	(t) helix with variable pitch and constant radius
	(u) straight line

Exercise-2

Marked Questions can be used as Revision Questions.

PART - I : ONLY ONE OPTION CORRECT TYPE

1. An α particle is moving along a circle of radius R with a constant angular velocity ω. Point A lies in the same plane at a distance 2R from the centre. Point A records magnetic field produced by α particle. If the minimum time interval between two successive times at which A records zero magnetic field is 't', the angular speed ω, in terms of t is -

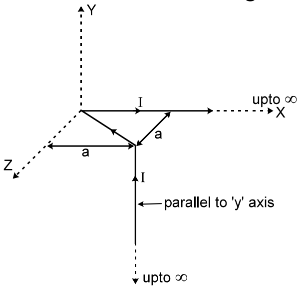
(A) $\frac{2\pi}{t}$ (B) $\frac{2\pi}{3t}$ (C) $\frac{\pi}{3t}$ (D) $\frac{\pi}{t}$

2. A particle is moving with velocity $\vec{v} = \hat{i} + 3\hat{j}$ and it produces an electric field at a point given by $\vec{E} = 2\hat{k}$. It will produce magnetic field at that point equal to (all quantities are in S.I. units) ($c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$)

(A) $\frac{6\hat{i} - 2\hat{j}}{c^2}$ (B) $\frac{6\hat{i} + 2\hat{j}}{c^2}$ (C) zero

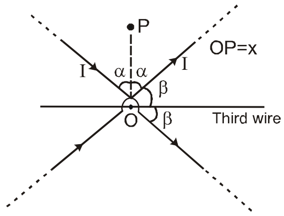
(D) can not be determined from the given data

3. The magnetic field at the origin due to the current flowing in the wire is



(A) $-\frac{\mu_0 I}{8\pi a}(\hat{i}+\hat{k})$ (B) $\frac{\mu_0 I}{2\pi a}(\hat{i}+\hat{k})$ (C) $\frac{\mu_0 I}{8\pi a}(-\hat{i}+\hat{k})$ (D) $\frac{\mu_0 I}{4\pi a\sqrt{2}}(\hat{i}-\hat{k})$

4. Three infinite current carrying conductors are placed as shown in figure. Two wires carry same current while current in third wire is unknown. The three wires do not intersect with each other and all of them are in the plane of paper. Which of the following is correct about a point 'P' which is also in the same lane :



(A) Magnetic field intensity at P is zero for all values of x.

(B) If the current in the third wire is $\frac{2I}{\sin\alpha}$ (left to right) then magnetic field will be zero at P for all values of x.

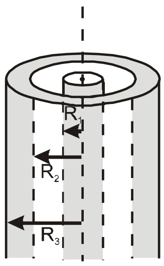
(C) If the current in the third wire is $\frac{2I}{\sin\alpha}$ (right to left) then magnetic field will be zero at P for all values of x.

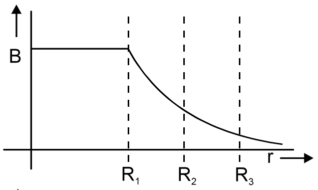
(D) None of these

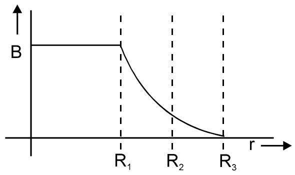
5. A long straight wire along the z – axis carries a current I in the negative Z direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x, y) in the z = 0 plane is:

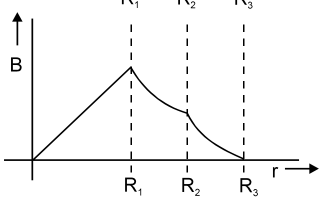
(A) $\frac{\mu_0 I}{2\pi}\frac{(y\hat{i}-x\hat{j})}{(x^2+y^2)}$ (B) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{i}+y\hat{j})}{(x^2+y^2)}$ (C) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{j}-y\hat{i})}{(x^2+y^2)}$ (D) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{i}-y\hat{j})}{(x^2+y^2)}$

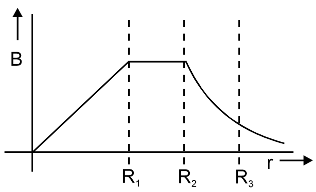
6. A coaxial cable is made up of two conductors. The inner conductor is solid and is of radius R_1 & the outer conductor is hollow of inner radius R_2 and outer radius R_3. The space between the conductors is filled with air. The inner and outer conductors are carrying currents of equal magnitudes and in opposite directions. Then the variation of magnetic field with distance from the axis is best plotted as:



(A)


(B)


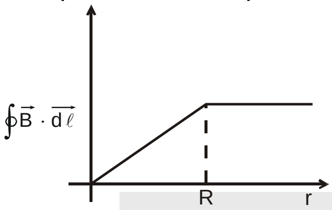
(C)


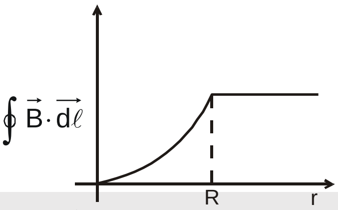
(D)


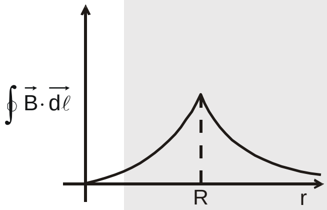
7. Axis of a solid cylinder of infinite length and radius R lies along y-axis it carries a uniformly distributed current 'i' along +y direction. Magnetic field at a point $\left(\frac{R}{2},\ y,\ \frac{R}{2}\right)$ is :-

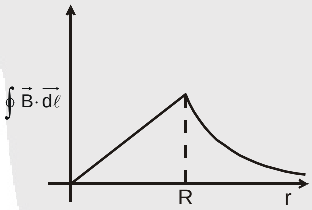
(A) $\frac{\mu_0 i}{4\pi R}(\hat{i}-\hat{k})$ (B) $\frac{\mu_0 i}{2\pi R}(\hat{j}-\hat{k})$ (C) $\frac{\mu_0 i}{4\pi R}\hat{j}$ (D) $\frac{\mu_0 i}{4\pi R}(\hat{i}+\hat{k})$

8. A cylindrical wire of radius R is carrying current i uniformly distributed over its cross-section. If a circular loop of radius 'r' is taken as amperian loop, then the variation value of $\oint \vec{B}.d\vec{\ell}$ over this loop with radius 'r' of loop will be best represented by:

(A) 

(B) 

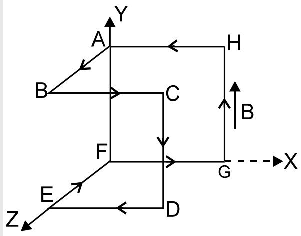
(C) 

(D) 

9. A constant direct current of uniform density $\vec{j}$ is flowing in an infinitely long cylindrical conductor. The conductor contains an infinitely long cylindrical cavity whose axis is parallel to that of the conductor and is at a distance $\vec{\ell}$ from it. What will be the magnetic induction $\vec{B}$ at a point inside the cavity at a distance $\vec{r}$ from the centre of cavity?

(A) $\frac{\mu_0(\vec{j}\times\vec{r})}{2}$ (B) $\frac{\mu_0(\vec{j}\times\vec{\ell})}{2}$ (C) $\frac{\mu_0(\vec{j}\times\vec{\ell})+\mu_0(\vec{j}\times\vec{r})}{2}$ (D) $\frac{\mu_0(\vec{j}\times\vec{\ell})-\mu_0(\vec{j}\times\vec{r})}{2}$

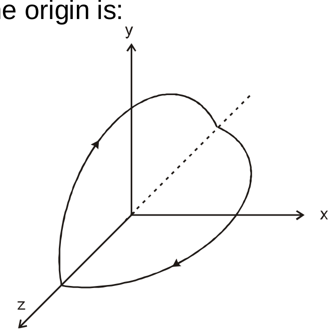
10. The given fig. shows a coil bent with AB = BC = CD = DE = EF = FG = GH = HA = 1 m and carrying current 1 A. There exists in space a vertical uniform magnetic field of 2 T in the y-direction. The torque on the loop is



(A) $2\hat{k}$ (B) $-2\hat{k}$

(C) $2\hat{j}$ (D) $4\hat{k}$

11. A non-planar circular loop consists of two semi-circles one of which lies in yz-plane & the other is in xz-plane as shown. The magnetic force experienced by positive charge of value Q moving with velocity v along x direction when it is at the origin is:

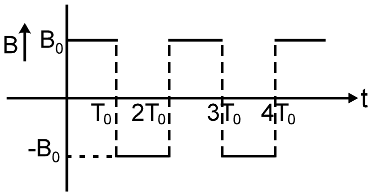


(A) $\frac{Qv\mu_0 I}{4R}$ (B) $\frac{Qv\mu_0 I}{2R}$ (C) $\frac{Qv\mu_0 I}{2\sqrt{2}R}$ (D) 0

12. A particle of mass ' m ' and charge ' q ' moves with a constant velocity ' v ' along the positive x direction. It enters a region containing a uniform magnetic field $\vec{B}$ directed along the negative Z direction, extending from x = a to x = b. The minimum value of v required so that the particle can just enter the region x > b is:

(A) $\frac{qbB}{m}$ (B) $\frac{q(b-a)B}{m}$ (C) $\frac{qaB}{m}$ (D) $\frac{q(b+a)B}{2m}$

13. In a region magnetic field along x axis changes with time according to the given graph.



If time period, pitch and radius of projected path are T_0, P_0 and R_0 respectively then which of the following is **incorrect** if the positively charged particle is projected at an angle θ_0 with the positive x-axis in x-y plane from origin:

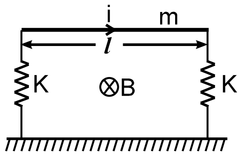
(A) At $t = \frac{T_0}{2}$, co-ordinates of charge are $\left(\frac{P_0}{2}, 0, -2R_0\right)$.

(B) At $t = \frac{3T_0}{2}$, co-ordinates of charge are $\left(\frac{3P_0}{2}, 0, 2R_0\right)$.

(C) The difference of maxismum and minimum z coordinate of the particle is $4R_0$

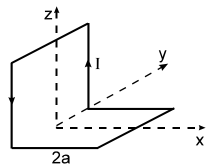
(D) The difference of maximum and minimum z coordinate of the particle is R_0

14. A horizontal metallic rod of mass ' m ' and length ' ℓ ' is supported by two vertical identical springs of spring constant ' K ' each and natural length ℓ_0. A current ' i ' is flowing in the rod in the direction shown. If the rod is in equilibrium then the length of each spring in this state is :



(A) $\ell_0 + \frac{i\ell B - mg}{K}$ (B) $\ell_0 + \frac{i\ell B - mg}{2K}$ (C) $\ell_0 + \frac{mg - i\ell B}{2K}$ (D) $\ell_0 + \frac{mg - i\ell B}{K}$

15. A non-planer loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length 2a. The magnetic field due to this loop at the point P(a, 0, a) points in the direction.



(A) $\frac{1}{\sqrt{2}}(-\hat{j}+\hat{k})$ (B) $\frac{1}{\sqrt{3}}(-\hat{j}+\hat{k}+\hat{i})$ (C) $\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$ (D) $\frac{1}{\sqrt{2}}(\hat{i}+\hat{k})$

16. A circular coil of radius R and a current I, which can rotate about a fixed axis passing through its diameter is initially placed such that its plane lies along magnetic field B. Kinetic energy of loop when it rotates through an angle 90° is : (Assume that I remains constant)

(A) $\pi R^2 BI$ (B) $\frac{\pi R^2 BI}{2}$ (C) $2\pi R^2 BI$ (D) $\frac{3}{2}\pi R^2 I$

17. A coil 2.0 cm in diameter has 300 turns. If the coil carries a current of 10 mA and lies in a magnetic field 5×10^{-2} T, the maximum torque experienced by the coil is :

(A) 4.7×10^{-2} N–m (B) 4.7×10^{-4} N–m (C) 4.7×10^{-5} N–m (D) 4.7×10^{-8} N–m

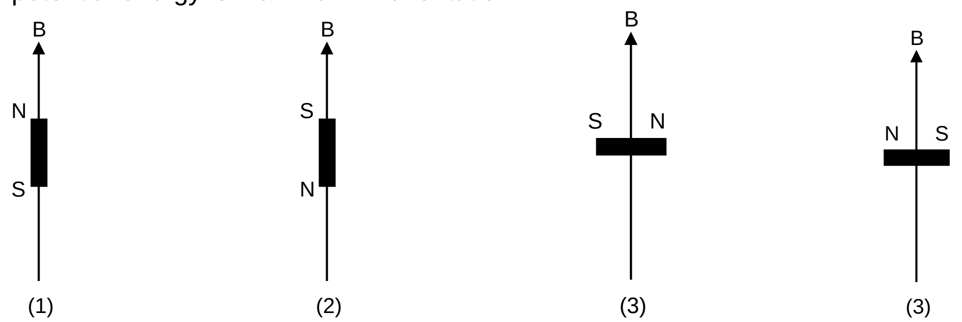
18. An infinitely long straight non-magnetic conducting wire of radius a carries a dc current I. The magnetic field B, at a distance r (r < a) from axis of the wire is :

(A) $\frac{\mu_0 I}{2\pi a}$ (B) $\frac{\mu_0 Ir}{2\pi a^2}$ (C) $\frac{2\mu_0 Ir}{\pi a^2}$ (D) $\frac{\mu_0 Ir^2}{2\pi a^3}$

19. The earth's magnetic field at a certain point is 7.0×10^{-5} T. This field is to be balanced by a magnetic field at the centre of a circular current carrying coil of radius 5.0 cm by suitably orienting it. If the coil has 100 turns then the required current is about

(A) 28 mA (B) 56 mA (C) 100 mA (D) 560 mA

20. Consider different orientations of a bar magnet lying in a uniform magnetic field as shown below. The potential energy is maximum in orientation



(A) 1 (B) 2 (C) 3 (D) 4

PART - II : NUMERICAL VALUE TYPE

1. A charge of 1 coulomb is placed at one end of a non-conducting rod of length 0.6m. The rod is rotated in a vertical plane about a horizontal axis passing through the other end of the rod with an angular velocity $10^4\pi$ rad/sec. Find the magnetic field at a point on the axis of rotation at a distance of 0.8m from the centre of the path is $N\pi \times 10^{-4}$T, then find value of N

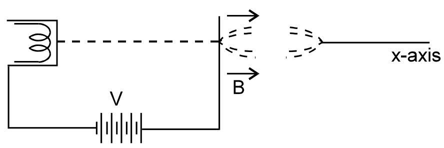
2. Magnetic field at the center of a rectangle of length 'ℓ' & width d, carrying a current i, is given by $\frac{N\mu_0 i}{\pi d}$, then find value of N (here $\ell >> d$)

3. A regular polygon of n sides is formed by bending a wire of length L which carries a current i. If the magnetic field B at the centre of the polygon $\frac{\mu_0 i n^2 \sin\frac{\pi}{n}\tan\frac{\pi}{n}}{N\pi L}$, then find the value of N.

4. A solid cylindrical conductor of radius R carries a current along its length. The current density J, however, is not uniform over the cross section of the conductor but is a function of the radius according to J = br, where b is a constant the magnetic field B. at $r_1 < R$ is $\frac{\mu_0 b r_1^2}{N}$. Then find value of N

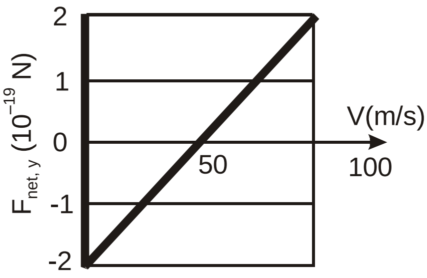
5. A capacitor of capacitance 50 μF is connected to a battery of 20 volts for a long time and then disconnected from it. It is now connected across a long solenoid having 8000 turns per metre. It is found that the potential difference across the capacitor drops to 90% of its maximum value in 2.0 seconds. The average magnetic field produced at the centre of the solenoid during this period is $N\pi \times 10^{-8}$ T. Then find value of N.

6. Electrons emitted with negligible speed from an electron gun are accelerated through a potential difference V along the X-axis. These electrons emerges from a narrow hole into a uniform magnetic field B directed along this axis. However, some of the electrons emerging from the hole make slightly divergent angles as shown in figure. These paraxial electrons meet for second time on the X-axis at a distance $\sqrt{\frac{N\pi^2 mV}{eB^2}}$. Then find value of N. (Neglect interaction between electrons)



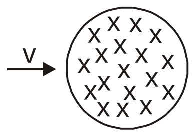
7. A particle of mass 1×10^{-26} kg and charge $+1.6 \times 10^{-19}$ C traveling with a velocity of 1.28×10^{6} m/s in the +x direction enters a region in which a uniform electric field E and a uniform magnetic field of induction B are present such that $E_x = E_y = 0$, $E_z = -102.4$ kV/m and $B_x = B_z = 0$, $B_y = 8 \times 10^{-2}$ Wb m^{-2}. The particle enters this region at the origin at time t = 0. If X -coordinate of the particle is $\frac{64}{N}$ at $t = 5 \times 10^{-6}$ s. Then find value of N.

8. At time t_1, an electron is sent along the positive direction of x-axis, through both an electric field $\vec{E}$ and a magnetic field $\vec{B}$, with $\vec{E}$ directed parallel to the y-axis. Graph gives the y-component $F_{net,y}$ of the net force on the electron due to the two fields, as a function of the electron's speed V at time t_1. Assuming $B_x = 0$, magnitude of electric field $\vec{E}$ in mN/C is 50X, then find X. (Use $e = 1.6 \times 10^{-19}$ C)



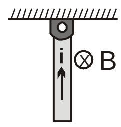
9. When a proton is released from rest in a gravity free room, it starts with an initial acceleration a_0 towards east. When it is projected towards south with a speed v_0, it moves with an initial acceleration $3a_0$ towards east. The minimum possible magnetic field in the room is $\frac{Nma_0}{ev_0}$. Find the value of N.

10. A uniform magnetic field exists in a circular region of radius r. The magnitude of magnetic field is B and points inward. An electron flies into the region radially as shown in the figure. After a certain time, the electron deflected by the magnetic field leaves the region. The time interval during which the electron moves in the region is $\frac{Nm\tan^{-1}\frac{eBr}{mV}}{eB}$. Then find value of N. (m is mass of electron and e is charge of electron)

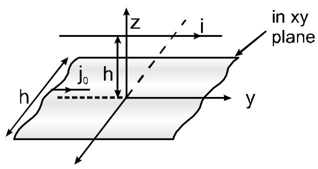


11. A long horizontal wire AB which is free to move in a vertical plane and carries a steady current of 20 A, is in equilibrium at a height of 0.01 m over another parallel infinitely long wire CD, which is fixed in a horizontal plane and carries a steady current of 30 A. Show that when AB is slightly depressed, it executes simple harmonic motion the period of oscillations is $\frac{1}{N}$ seconds, then find the value of N. (Use $\pi^2 = 10$ & $g = 10$m/s^2)

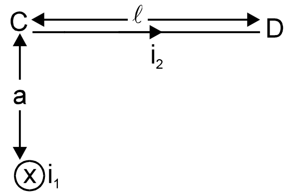
12. A uniform rod of length L and mass M is hinged at its upper point and is at rest at that moment in the vertical plane. A current i flows in it. A uniform magnetic field of strength B exists perpendicular to the rod and in horizontal direction (as shown). The force exerted by the hinge on the rod just after release in horizontal direction is $\frac{iLB}{N}$. Then find the value of N.



13. A conductor placed along the line z = h carrying a current i is placed parallel to a very long current strip of width ' h ' placed in the xy plane and carrying a current per unit width J_0 as shown in the figure. The force per unit length on the conductor is $\frac{\mu_0 iJ_0}{\pi}\tan^{-1}\left(\frac{1}{N}\right)(-\hat{k})$. The sheet is placed symmetrically with respect to y-axis, Then find value of N.



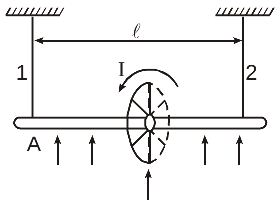
14. An infinitely long straight wire carries a current i_1, as shown in the figure. The total force on another wire CD of length ℓ which is placed so that c is at a distance a from the current carrying wire is $\frac{\mu_0 i_1 i_2}{N\pi}\ln\left(\frac{a^2+\ell^2}{a^2}\right)$. Then find value of N



15. Consider a nonconducting ring of radius r and mass m which has a total charge q distributed uniformly on it. The ring is rotated about its axis with an angular speed ω. (a) The equivalent electric current in the ring is $\frac{q\omega}{N\pi}$. Then find value of N (b) The magnetic moment of the ring is $\frac{4q\omega r^2}{N}$. Then find value of N.

16. Consider a nonconducting disc of radius r and mass m which has a charge q distributed uniformly over it. The disk is rotated about its axis with an angular speed ω. Magnetic moment of the disc is $\frac{1}{N}q\omega r^2$. Then find value of N.

17. The circular current loop of radius b = 2cm shown in the figure is mounted rigidly on the axle, midway between the two supporting cords. Current in the loop is 100 amp. The length of axle is ℓ = 1 m. Absence of an external magnetic field the tensions in the cords are equal to $T_0 = 10\pi$ N

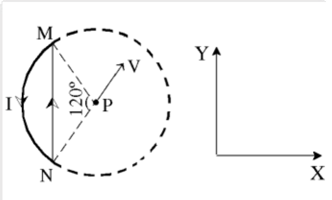


When the vertical magnetic field B = 100 T is present and the axle still in equilibrium, then the tensions (in newtons) in the cords are $T_1 = n\pi$; $T_2 = m\pi$. Find the magnitude of m – n

Note: Current in anticlockwise direction as seen from right side.

18. A wire loop of radius a carrying current I is placed in the X-Y plane as shown in the figure

(a) If a particle with charge +Q and mass m is placed at the centre p and given a velocity along NP (fig). Magnitude of its initial instantaneous acceleration acceleration is $\frac{QV}{m}\frac{\mu_0 I}{6a}\left(\frac{N\sqrt{3}}{\pi}-1\right)$, then find value of N

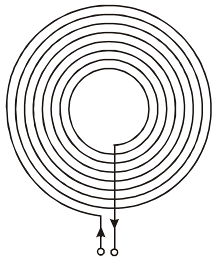


(b) If an external uniform magnetic induction field $\vec{B} = B\hat{i}$ is applied. The torque acting on the loop due to the field is $\vec{\tau} = BI\left(\frac{\pi}{3}-\frac{\sqrt{3}}{N}\right)a^2\ \hat{j}$. Then find value of N.

19. A thin insulated wire forms a plane spiral of N = 100 tight turns carrying a current I = 8 mA. The radii of inside and outside turns (figure) are equal to a = 50 mm and b = 100 mm. :

(a) The magnetic induction at the centre of the spiral is N μT; Then find value of N.

(b) The magnetic moment of the spiral with a given current is N mA.m^2. Then find value of N.

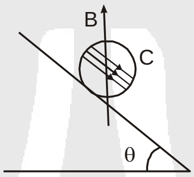


20. A square frame carrying a current $I = 0.90A$ is located in the same plane as a long straight wire carrying a current $I_0 = 5.0$ A. The frame side has a length $a = 8.0$ cm. The axis of the frame passing through the midpoints of opposite sides is parallel to the wire and is separated from it by the distance which is $\eta = 1.5$ times greater than the side of the frame.

(a) Ampere force (magnetic force) acting on the frame is $\frac{9}{P}$ μN ; Then find value of P.

(b) The magnitude of mechanical work is $(144\ell nP) \times 10^{-9}$ J to be performed in order to turn the frame slowly through 180° about its axis, with the currents maintained constant. Then find value of P

21. Figure shows (only cross section) a wooden cylinder C with a mass m of 0.25 kg, a radius R and a length ℓ perpendicular to the plane of paper of 0.1 meter with N = 10 where N is number of turns of wire wrapped around it longitudinally, so that the plane of the wire loop contains the axis of the cylinder. The least current through the loop is $\frac{P}{2}$ A that will prevent the cylinder from moving down a plane whose surface is inclined at angle θ to the horizontal, in the presence of a vertical field of magnetic induction 0.5 weber/meter^2, if the plane of the windings is parallel to the inclined plane? (Bottom most point of cylinder does not slip) then find value of P.

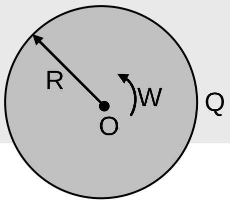


PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

1. A magnetic needle (small magnet) is kept in a nonuniform magnetic field. It
(A) may experience a force and torque
(B) may experience a force but not a torque
(C) may experience a torque but not a force
(D) will experience neither a force nor a torque

2. The magnetic field at the origin due to a current element $i\, d\vec{\ell}$ placed at a position $\vec{r}$ is
(A) $\frac{\mu_0 i}{4\pi}\frac{d\vec{l}\times\vec{r}}{r^3}$
(B) $-\frac{\mu_0 i}{4\pi}\frac{\vec{r}\times d\vec{\ell}}{r^3}$
(C) $\frac{\mu_0 i}{4\pi}\frac{\vec{r}\times d\vec{\ell}}{r^3}$
(D) $-\frac{\mu_0 i}{4\pi}\frac{d\vec{\ell}\times\vec{r}}{r^3}$

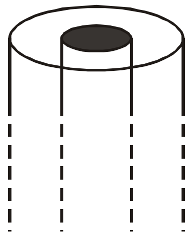
3. A non-conducting disc having uniform positive charge Q, is rotating about its axis with uniform angular velocity ω. The magnetic field at the center of the disc is.



(A) directed outward
(B) having magnitude $\frac{\mu_0 Q\omega}{4\pi R}$
(C) directed inwards
(D) having magnitude $\frac{\mu_0 Q\omega}{2\pi R}$

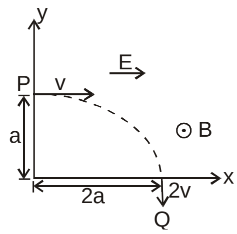
4. A hollow tube is carrying an electric current along its length distributed uniformly over its surface. The magnetic field
(A) Increases linearly from the axis to the surface
(B) is constant inside the tube
(C) is zero at the axis
(D) is non-zero outside the tube at finite distance from surface

5. In a coaxial, straight cable, the central conductor and the outer conductor carry equal currents in opposite directions. The magnetic field is non-zero.



(A) outside the cable
(B) inside the inner conductor except axis of the conductor
(C) all the point inside the outer conductor and outside inner conductor
(D) only on axis of conductors.

6. A particle of charge +q and mass m moving under the influence of a uniform electric field E and a uniform magnetic field $B\hat{k}$ follows a trajectory from P and Q as shown in figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$. Which of the following statement(s) is/are correct?



(A) $E = \frac{3}{4}\left(\frac{mv^2}{qa}\right)$

(B) Rate of work done by the electric field at P is $\frac{3}{4}\left(\frac{mv^3}{a}\right)$

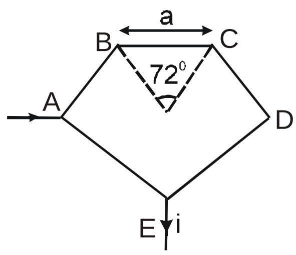
(C) Rate of work done by the electric field at P is zero.
(D) Rate of work done by both fields at Q is zero.

7. Let $\vec{E}$ and $\vec{B}$ denote the electric and magnetic fields in a certain region of space. A proton moving with a velocity along a straight line enters the region and is found to pass through it undeflected. Indicate which of the following statements are possible for the observation :

(A) $\vec{E} = 0$ and $\vec{B} = 0$ (B) $\vec{E} \neq 0$ and $\vec{B} = 0$

(C) $\vec{E} \neq 0$, $\vec{B} \neq 0$ and both $\vec{E}$ and $\vec{B}$ are parallel to $\vec{v}$ (D) $\vec{E}$ is parallel to $\vec{v}$ but $\vec{B}$ is perpendicular to $\vec{v}$

8. Magnetic field strength at the centre of regular pentagon made of a conducting wire of uniform cross section area as shown in figure is :



(A) $\frac{5\mu_0 i}{4\pi a}\left[2\sin\frac{72°}{2}\right]$ (B) 0

(C) Not zero if current 'i' leaves D point instead of E
(D) Zero even if the current 'i' leaves point D instead of point E

9. H^+, He^+ and O^{2+} all having the same kinetic energy pass through a thin region in which there is a uniform magnetic field perpendicular to their velocity. The masses of H^+, He^+ and O^{2+} are 1 amu, 4amu and 16 amu respectively, then
(A) H^+ will be deflected most
(B) O^{2+} will be deflected most
(C) He^+ and O^{2+} will be deflected equally
(D) All will be deflected equally

10. A beam of electrons moving with a momentum p enters a uniform magnetic field of flux density B perpendicular to its motion. Which of the following statement(s) is (are) true?

(A) Energy gained is $\frac{p^2}{2m}$ (B) Centripetal force on the electron is $Be\frac{m}{p}$

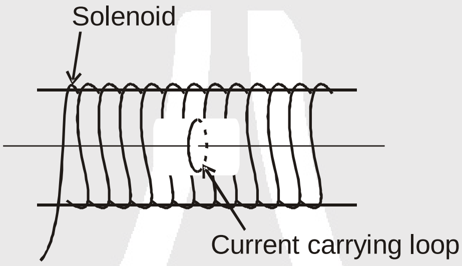
(C) Radius of the electron's path is $\frac{p}{Be}$ (D) Work done on the electrons by the magnetic field is zero

11. Two ions have equal masses but one is singly-ionized and other is tripply-ionized. They are projected from the same place in a uniform magnetic field with the same velocity perpendicular to the field.
(A) Both ions will go along circles of equal radii.
(B) The circle described by the single-ionized charge will have a radius tripply that of the other circle
(C) The two circles do not touch each other
(D) The two circles touch each other

12. A charged particle goes undeflected in a region containing uniform electric and magnetic field. It is not possible at all ($\vec{\upsilon}$ = velocity of particle, $\vec{E}$ = uniform electric field, $\vec{B}$ = uniform magnetic field)
(A) $\vec{E} \parallel \vec{B}$, $\vec{\upsilon} \parallel \vec{E}$
(B) $\vec{E}$ is not collinear to $\vec{B}$
(C) $\vec{\upsilon} \parallel \vec{B}$ but $\vec{E}$ is not collinear to $\vec{B}$
(D) $\vec{E} \parallel \vec{B}$ but $\vec{\upsilon}$ is not collinear to $\vec{E}$

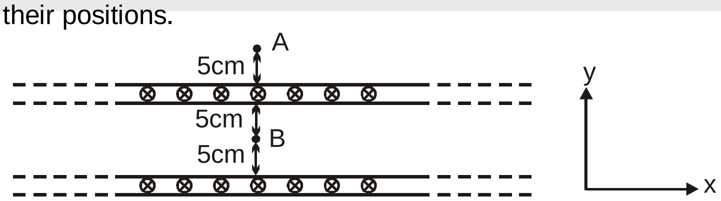
13. A charged particle moves along a circle under the action of possible uniform & constant electric and magnetic fields. Which of the following are not possible at all ?
(A) E = 0, B = 0 (B) E = 0, B $\neq$ 0 (C) E $\neq$ 0, B = 0 (D) E $\neq$ 0, B $\neq$ 0

14. A single circular loop of wire with radius 0.02 cm carries a current of 8.0 A. It is placed at the centre of a solenoid that has length 0.65 m, radius 0.080 m and 1300 turns.



(A) The value of the current in the solenoid so that the magnetic field at the centre of the loop becomes zero, is equal to 4.4 A.
(B) The value of the current in the solenoid so that the magnetic field at the centre of the loop becomes zero, is equal to 10 A.
(C) The magnitude of the total magnetic field at the centre of the loop (due to both the loop and the solenoid) if the current in the loop is reversed in direction from that needed to make the total field equal to zero T esla, is $8\pi \times 10^{-3}$ T.
(D) The magnitude of the total magnetic field at the centre of the loop (due to both the loop and the solenoid) if the current in the loop is reversed in direction from that needed to make the total field equal to zero Tesla, is $16\pi \times 10^{-3}$ T.

15. Figure shows cross section of two large parallel metal sheets carrying electric currents along their surfaces. The current in each sheet is $\frac{10}{\pi}$ A/m along the width. Consider two points A and B, as shown in the figure with their positions.



(A) Magnetic field at A is 4μT along x-direction.
(B) Magnetic field at A is 4μT along negative x-direction.
(C) Magnetic field at B is zero.
(D) Magnetic field at B is 2μT along x-direction.

16. Let $[\in_0]$ denote the dimensional formula of the permittivity of the vacuum and $[\mu_0]$ that of the permeability of the vacuum. If M = mass, L = length, T = time and I = electric current,
(A) $[\in_0] = M^{-1}L^{-3}T^2I$ (B) $[\in_0] = M^{-1}L^{-3}T^4I^2$ (C) $[\mu_0] = MLT^{-2}I^{-2}$ (D) $[\mu_0] = ML^2T^{-1}I$

17. A positively charged particle is moving along the positive X-axis. You want to apply a magnetic field for a short time so that the particle may reverse its direction and move parallel to the negative X-axis. This can be done by applying the magnetic field along.
(A) Y-axis (B) Z-axis (C) Y-axis only (D) Z-axis only.

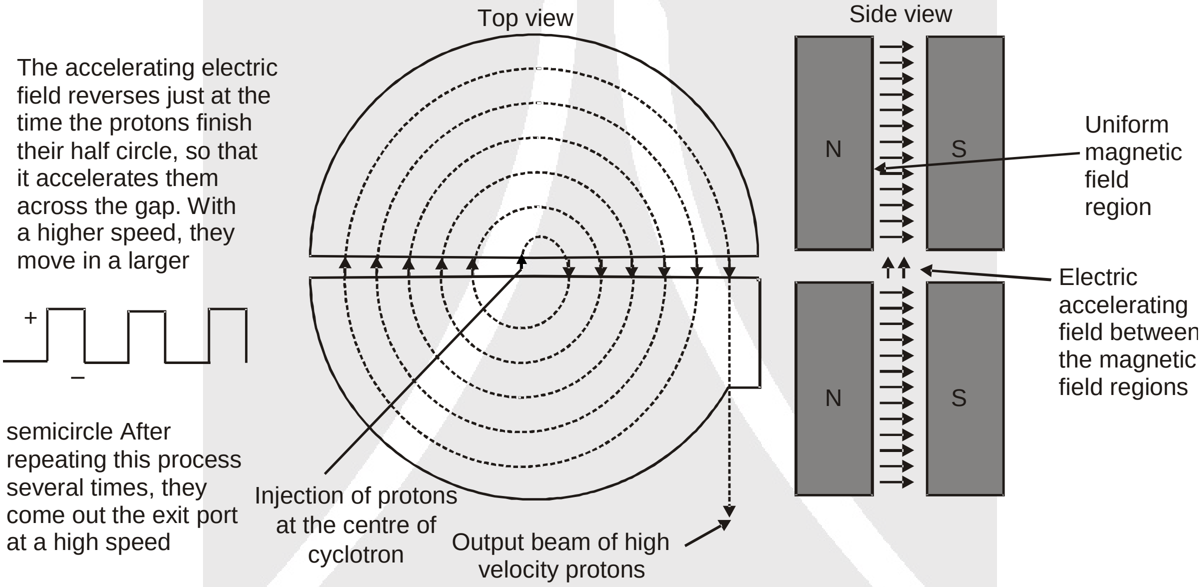
18. A small bar magnet is suspended by a thread. A torque is applied and the magnet is found to execute angular oscillations. The time period of oscillations
(A) decreases with the moment of the magnet
(B) increases with the increase of the horizontal component of the earth's magnetic filed
(C) will remain unchanged even if another magnet is kept at a distance
(D) depends on the mass of the magnet

PART - IV : COMPREHENSION

Comprehension # 1

(Read the following passage and answer the questions. They have only one correct option)

In the given figure of a cyclotron, showing the particle source S and the dees. A uniform magnetic field is directed up from the plane of the page. Circulating protons spiral outward within the hollow dees, gaining energy every time they cross the gap between the dees.



Suppose that a proton, injected by source S at the centre of the cyclotron in figure initially moves toward a negatively charged dee. It will accelerate toward this dee and enter it. Once inside, it is shielded from electric field by the copper walls of the dee; that is the electric field does not enter the dee. The magnetic field, however, is not screened by the (nonmagnetic) copper dee, so the proton moves in circular path whose radius, which depends on its speed, is given by

$$r = \frac{mv}{qB} \qquad ...(1)$$

Let us assume that at the instant the proton emerges into the center gap from the first dee, the potential difference between the dees is reversed. Thus, the proton again faces a negatively charged dee and is again accelerated. Thus, the proton again faces a negatively charged dee and is again accelerated. This process continues, the circulating proton always being in step, with the oscillations of the dee potential, until the proton has spiraled out to the edge of the dee system. There a deflector plate sends it out through a portal.

The key to the operation of the cyclotron is that the frequency f at which the proton circulates in the field (and that does not depend on its speed) must be equal to the fixed frequency f_{osc} of the electrical oscillator, or

$$f = f_{osc} \text{ (resonance condition).} \qquad ...(2)$$

This resonance condition says that, if the energy of the circulating proton is to increase, energy must be fed to it at a frequency f_{osc} that is equal to the natural frequency f at which the proton circulates in the magnetic field.

Combining equation 1 and 2 allows us to write the resonance condition as

$$qB = 2\pi m f_{osc}. \qquad ...(3)$$

For the proton, q and m are fixed. The oscillator (we assume) is designed to work at a single fixed frequency f_{osc}. We then "tune" the cyclotron by varying B until eq. 3 is satisfied and then many protons circulate through the magnetic field, to emerge as a beam.

1. Ratio of radius of successive semi circular path

(A) $\sqrt{1} : \sqrt{2} : \sqrt{3} : \sqrt{4}$ .... (B) $\sqrt{1} : \sqrt{3} : \sqrt{5}$ ....

(C) $\sqrt{2} : \sqrt{4}\ \sqrt{6}$: .... (D) 1 : 2 : 3 ....

2. Change in kinetic energy of charge particle after every time period is :

(A) 2qV (B) qV (C) 3qV (D) None of these

3. If q/m for a charge particle is 10^6, frequency of applied AC is 10^6 Hz. Then applied magnetic field is:

(A) 2π tesla (B) π tesla (C) 2 tesla (D) can not be defined

4. Distance travelled in each time period are in the ratio of :

(A) $\sqrt{1}+\sqrt{3} : \sqrt{5}+\sqrt{7} : \sqrt{9}+\sqrt{11}$ (B) $\sqrt{2} + \sqrt{3} : \sqrt{4} + \sqrt{5} : \sqrt{6} + \sqrt{7}$

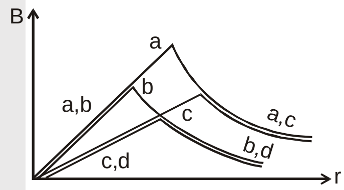
(C) $\sqrt{1} : \sqrt{2} : \sqrt{3}$ (D) $\sqrt{2} : \sqrt{3} : \sqrt{4}$

5. For a given charge particle a cyclotron can be "tune" by :

(A) changing applied A.C. voltage only
(B) changing applied A.C. voltage and magnetic field both
(C) changing applied magnetic field only
(D) by changing frequency of applied A.C.

Comprehension # 2

Curves in the graph shown give the magnitude B of the magnetic field (due to individual wire) inside and outside four long wires a, b, c and d, carrying currents that are uniformly distributed across the cross sections of the wires as functions of radial distance r (from the axis). Overlapping portions of the plots are indicated by double labels. All curves start from the origin.



6. Which wire has the greatest radius ?

(A) a (B) b (C) c (D) d

7. Which wire has the greatest magnitude of the magnetic field on the surface?

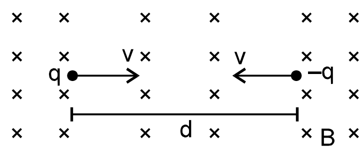
(A) a (B) b (C) c (D) d

8. The current density in wire a is

(A) greater than in wire c. (B) less than in wire c.
(C) equl to that in wire c. (D) not comparable to that of in wire c due to lack of information.

Comprehension # 3

Two particles, each having a mass m are placed at a separation d in a uniform magnetic field B as shown in figure. They have opposite charges of equal magnitude q. At time t = 0, the particles are projected towards each other, each with a speed v. (Neglect interaction between two charges)



9. Find the maximum value v_m of the projection speed so that the two particles do not collide.

(A) $\frac{qBd}{4m}$ (B) $\frac{2qBd}{m}$ (C) $\frac{qBd}{m}$ (D) $\frac{qBd}{2m}$

10. What would be the minimum separation between the particles if $v = v_m/4$?

(A) $\frac{d}{4}$ (B) $\frac{3d}{5}$ (C) $\frac{3d}{4}$ (D) $\frac{5d}{4}$

11. What would be the maximum separation between the particles if $v = v_m/4$?

(A) $\frac{5d}{2}$ (B) $\frac{5d}{4}$ (C) $\frac{d}{4}$ (D) $\frac{3d}{4}$

12. At what instant will a collision occur between the particles if $v = 2v_m$?

(A) $\frac{\pi m}{6qB}$ (B) $\frac{\pi m}{qB}$ (C) $\frac{6\pi m}{qB}$ (D) $\frac{\pi m}{2qB}$

13. Suppose $v = 2v_m$ and they stick to each other after the collision. Describe the motion after the collision (neglect the magnetic, gravitational & electric forces between charges).

(A) circular motion (B) parabolic motion (C) elliptical motion (D) straight line

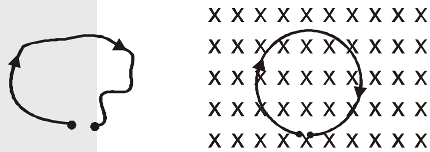
Exercise-3

✍ Marked Questions can be used as Revision Questions.

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

*** Marked Questions may have more than one correct option.**

1. A thin flexible wire of length L is connected to two adjacent fixed points carries a current I in the clockwise direction, as shown in the figure. When system is put in a uniform magnetic field of strength B going into the plane of paper, the wire takes the shape of a circle. The tension in the wire is : **[JEE 2010, 3/163, –1]**

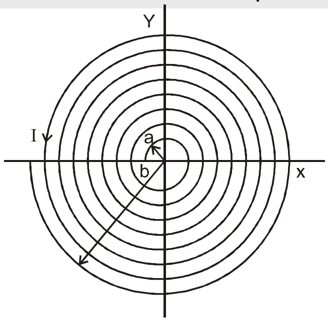


(A) IBL (B) $\frac{IBL}{\pi}$ (C) $\frac{IBL}{2\pi}$ (D) $\frac{IBL}{4\pi}$

2. An electron and a proton are moving on straight parallel paths with same velocity. They enter a semi-infinite region of uniform magnetic field perpendicular to the velocity. Which of the following statement(s) is/are true? **[JEE 2011, 4/160]**

(A) They will never come out of the magnetic field region.
(B) They will come out travelling along parallel paths.
(C) They will come out at the same time.
(D) They will come out at different times.

3. A long insulated copper wire is closely wound as a spiral of *'N'* turns. The spiral has inner radius 'a' and outer radius *'b'*. The spiral lies in the X-Y plane and a steady current I flows through the wire. The Z-component of the magnetic field at the center of the spiral is **[JEE - 2011' 3/160, –1]**



(A) $\frac{\mu_0 NI}{2(b-a)}\ell n\left(\frac{b}{a}\right)$ (B) $\frac{\mu_0 NI}{2(b-a)}\ell n\left(\frac{b+a}{b-a}\right)$ (C) $\frac{\mu_0 NI}{2b}\ell n\left(\frac{b}{a}\right)$ (D) $\frac{\mu_0 NI}{2b}\ell n\left(\frac{b+a}{b-a}\right)$

4. Consider the motion of a positive point charge in a region where there are simultaneous uniform electric and magnetic fields $\vec{E} = E_0\hat{j}$ and $\vec{B} = B_0\hat{j}$. At time t = 0, this charge has velocity $\vec{v}$ in the x-y plane, making an angle θ with x-axis. Which of the following option(s) is(are) correct for time t > 0 ?

(A) If θ = 0°, the charge moves in a circular path in the x-z plane.

(B) If θ = 0°, the charge undergoes helical motion with constant pitch along the y-axis.

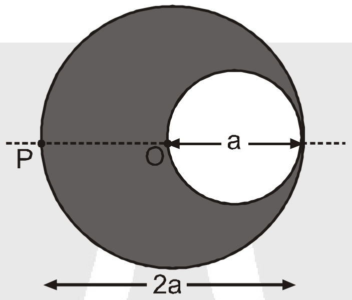
(C) If θ = 10°, the charge undergoes helical motion with its pitch increasing with time, along the y-axis.

(D) If θ = 90°, the charge undergoes linear but accelerated motion along the y-axis.

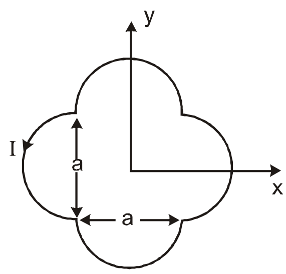
[IIT-JEE-2012, Paper-1; 4/70]

5. A cylindrical cavity of diameter a exists inside a cylinder of diameter 2a shown in the figure. Both the cylinder and the cavity are infinitely long. A uniform current density J flows along the length. If the magnitude of the magnetic field at the point P is given by $\frac{N}{12}\mu_0 aJ$, then the value of N is :

[IIT-JEE-2012, Paper-1; 4/70]



6. A loop carrying current I lies in the x-y plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. the magnetic moment of the current loop is : **[IIT-JEE-2012, Paper-2; 3/66, –1]**



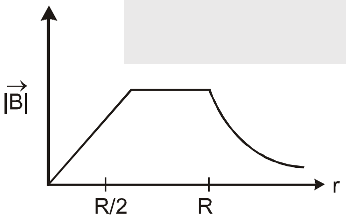
(A) $a^2 I\hat{k}$

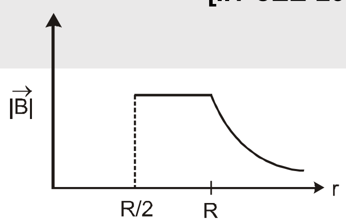
(B) $\left(\frac{\pi}{2}+1\right)a^2 I\hat{k}$

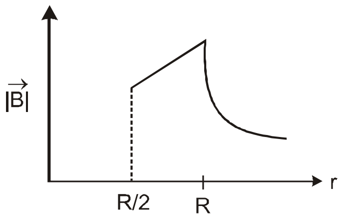
(C) $-\left(\frac{\pi}{2}+1\right)a^2 I\hat{k}$

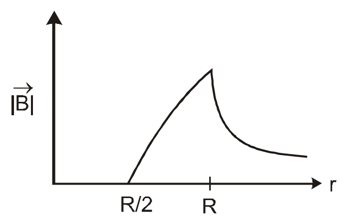
(D) $(2\pi+1)a^2 I\hat{k}$

7. An infinitely long hollow conducting cylinder with inner radius R/2 and outer radius R carries a uniform current density along is length. The magnitude of the magnetic field, $|\vec{B}|$ as a function of the radial distance r from the axis is best represented by : **[IIT-JEE-2012, Paper-2; 3/66, –1]**

(A) 

(B) 

(C) 

(D) 

8. A particle of mass M and positive charge Q, moving with a constant velocity $\vec{u}_1 = 4\hat{i}ms^{-1}$, enters a region of uniform static magnetic field normal to the x-y plane. The region of the magnetic field extends from x = 0 to x = L for all values of y. After passing through this region, the particle emerges on the other side after 10 milliseconds with a velocity $\vec{u}_2 = 2\left(\sqrt{3}\hat{i} + \hat{j}\right)ms^{-1}$. The correct statement(s) is (are) :

[JEE (Advanced)-2013, 4/60, –1]

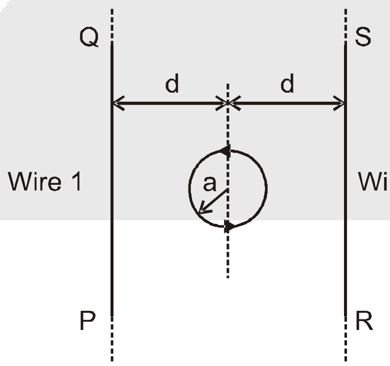
(A) The direction of the magnetic field is –z direction.
(B) The direction of the magnetic field is +z direction
(C) The magnitude of the magnetic field $\frac{50\pi M}{3Q}$ units.
(D) The magnitude of the magnetic field is $\frac{100\pi M}{3Q}$ units.

9. A steady current I flows along an infinitely long hollow cylindrical conductor of radius R. This cylinder is placed coaxially inside an infinite solenoid of radius 2R. The solenoid has n turns per unit length and carries a steady current I. Consider a point P at a distance r from the common axis. The correct statement(s) is (are) : **[JEE (Advanced)-2013, 3/60, –1]**
(A) In the region 0 < r < R, the magnetic field is non–zero.
(B) In the region R < r < 2R, the magnetic field is along the common axis.
(C) In the region R < r < 2R, the magnetic field is tangential to the circle of radius r, centered on the axis.
(D) In the region r > 2R, the magnetic field is non–zero.

10. Two parallel wires in the plane of the paper are distance X_0 apart. A point charge is moving with speed u between the wires in the same plane at a distance X_1 from one of the wires. When the wires carry current of magnitude I in the same direction, the radius of curvature of the path of the point charge is R_1. In contrast, if the currents I in the two wires have direction opposite to each other, the radius of curvature of the path is R_2. If $\frac{x_0}{x_1}$ = 3, the value of $\frac{R_1}{R_2}$ is : **[JEE (Advanced)-2014,P-1, 3/60]**

Paragraph For Questions 11 to 12

The figure shows a circular loop of radius a with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the centre of the loop is d. The loop and the wires are carrying the same current I. The current in the loop is in the counterclockwise direction if seen from above. **[JEE (Advanced) 2014, P-2, 3/60, –1]**



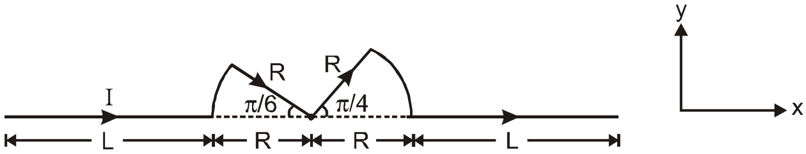
11. When d ≈ a but wires are not touching the loop, it is found that the net magnetic filed on the axis of the loop is zero at a height h above the loop. In that case **[JEE (Advanced)-2014, 3/60, –1]**
(A) current in wire 1 and wire 2 is the direction PQ and RS, respectively and h ≈ a
(B) current in wire 1 and wire 2 is the direction PQ and SR, respectively and h ≈ a
(C) current in wire 1 and wire 2 is the direction PQ and SR, respectively and h ≈ 1.2 a
(D) current in wire 1 and wire 2 is the direction PQ and RS, resepectively and h ≈ 1.2 a

12. Consider d >> a, and the loop is rotated about its diameter parallel to the wires by 30º from the position shown in the figure. If the currents in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop)

[JEE (Advanced)-2014, 3/60, –1]

(A) $\frac{\mu_0 I^2 a^2}{d}$ (B) $\frac{\mu_0 I^2 a^2}{2d}$ (C) $\frac{\sqrt{3}\mu_0 I^2 a^2}{d}$ (D) $\frac{\sqrt{3}\mu_0 I^2 a^2}{2d}$

13. A conductor (shown in the figure) carrying contant current I is kept in the x-y plane in a uniform magnetic field $\vec{B}$. IF F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is (are) : **[JEE(Advanced) 2015 ; 4/88, –2]**



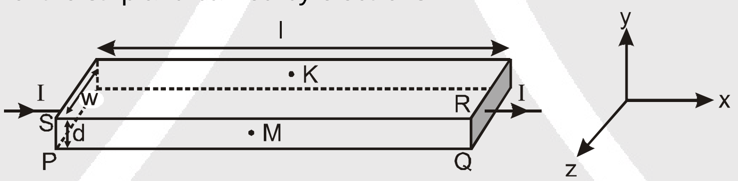
(A) If $\vec{B}$ is along $\hat{z}$, $F \propto (L + R)$ (B) If $\vec{B}$ is along $\hat{x}$, $F = 0$

(C) If $\vec{B}$ is along $\hat{y}$, $F \propto (L + R)$ (D) If $\vec{B}$ is along $\hat{z}$, $F = 0$

Paragraph For Questions 14 to 15

In a thin rectangular metallic strip a constant current I flows along the positive x-direction, as shown in the figure. The length, width and thicknes of the strip are l, w and d, respectively.

A uniform magnetic field $\vec{B}$ is applied on the strip along the positive y-direction. Due to this, the charge carriers experience a net deflection along the z-direction. This results in accumulation of charge carriers on the surface PQRS and is appearance of equal and opposite charges on the face opposite to PQRS. A potential difference along the z-direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross section of the strip and carried by electrons.



14. Consider two different metallic strips (1 and 2) fo the same material. Their lengths are the same, width are w_1 and w_2 and thicknesses are d_1 and d_2, respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x-y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B, the correct statement(s) is (are)

[JEE(Advanced) 2015 ; P-2,4/88, –2]

(A) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$ (B) If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = V_1$

(C) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = 2V_1$ (D) If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = V_1$

15. Consider two different metallic strips (1 and 2) of same dimensions (length l, width w and thickness d) with carrier densities n_1 and n_2, respectively. Strip 1 is placed in magnetic field B_1 and strip 2 is placed in magnetic field B_2, both along positive y-directions. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is (are) **[JEE(Advanced) 2015 ; P-2,4/88, –2]**

(A) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$ (B) If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$

(C) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5V_1$ (D) If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$

Answer Q.16. Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

A charged particle (electron or proton) is introduced at the origin (x = 0, y = 0, z = 0) with a given initial velocity $\vec{v}$. A uniform electric field $\vec{E}$ and a uniform magnetic field $\vec{B}$ exist everywhere. The velocity $\vec{v}$, electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in column 1,2 and 3, respectively. The quantities E_0, B_0 are positive in magnitude.

	Column -1		Column -2		Column -3
(I)	Electron with $\vec{v} = 2\frac{E_0}{B_0}\hat{x}$	(i)	$\vec{E} = E_0\hat{z}$	(P)	$\vec{B} = -B_0\hat{x}$
(II)	Electron with $\vec{v} = \frac{E_0}{B_0}\hat{y}$	(ii)	$\vec{E} = -E_0\hat{y}$	(Q)	$\vec{B} = B_0\hat{x}$
(III)	Proton with $\vec{v} = 0$	(iii)	$\vec{E} = -E_0\hat{x}$	(R)	$\vec{B} = B_0\hat{y}$
(IV)	Proton with $\vec{v} = 2\frac{E_0}{B_0}\hat{x}$	(iv)	$\vec{E} = E_0\hat{x}$	(S)	$\vec{B} = B_0\hat{z}$

16. In which case will the particle move in a straight line with constant velocity ?

[JEE(Advanced) 2017 ; P-1, 3/61, –1]

(A) (IV) (i) (S) (B) (III) (ii) (R) (C) (II) (iii) (S) (D) (III) (iii) (P)

17. In which case will the particle describe a helical path with axis along the positive z direction ?

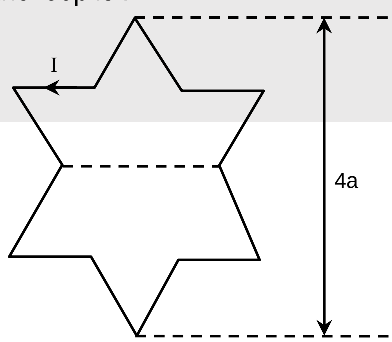
[JEE(Advanced) 2017 ; P-1, 3/61, –1]

(A) (IV) (i) (S) (B) (II) (ii) (R) (C) (III) (iii) (P) (D) (IV) (ii) (R)

18. In which case would the particle move in a straight line along the negative direction of y-axis (i.e, move along $-\hat{y}$) ? **[JEE(Advanced) 2017 ; P-1, 3/61, –1]**

(A) (III) (ii) (R) (B) (IV) (ii) (S) (C) (III) (ii) (P) (D) (II) (iii) (Q)

19. A symmetric star shaped conducting wire loop is carrying a steady state current I as shown in the figure. The distance between the diametrically opposite vertices of the star is 4a. The magnitude of the magnetic field at the center of the loop is : **[JEE(Advanced) 2017 ; P-2, 3/61, –1]**

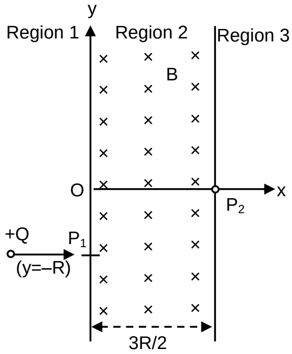


(A) $\frac{\mu_0 I}{4\pi a}6\left[\sqrt{3}-1\right]$ (B) $\frac{\mu_0 I}{4\pi a}6\left[\sqrt{3}+1\right]$

(C) $\frac{\mu_0 I}{4\pi a}3\left[2-\sqrt{3}\right]$ (D) $\frac{\mu_0 I}{4\pi a}3\left[\sqrt{3}-1\right]$

20*. A uniform magnetic field B exists in the region between x = 0 and x = $\frac{3R}{2}$ (region 2 in the figure) pointing normally into the plane of the paper. A particle with charge +Q and momentum p directed along x-axis enters region 2 from region 1 at point P_1 (y = –R). Which of the following option(s) is/are correct ?

[JEE (Advanced) 2017 ; P-2, 4/61, –2]



(A) When the particle re-enters region 1 through the longest possible path in region 2, the magnitude of the change in its linear momentum between P_1 and the farthest point from y-axis is $p/\sqrt{2}$.

(B) For a fixed B, particles of same charge Q and same velocity v, the distance between the point P_1 and the point of re-entry into region 1 is inversely proportional to the mass of the particle.

(C) For $B = \frac{8}{13}\frac{p}{QR}$, the particle will enter region 3 through the point P_2 on x-axis

(D) For $B > \frac{2}{3}\frac{p}{QR}$, the particle will re-enter region 1.

21*. Two infinitely long straight wires lie in the xy-plane along the lines x = ±R. The wire located at x = +R carries a constant current I_1 and the wire located at x = – R carries a constant current I_2. A circular loop of radius R is suspended with its centre at (0, 0, $\sqrt{3}R$) and in a plane parallel to the xy-plane. This loop carries a constant current I in the clockwise direction as seen from above the loop. The current in the wire is taken to be positive if it is in the $+\hat{j}$ direction. Which of the following statements regarding the magnetic field $\vec{B}$ is (are) true? **[JEE (Advanced) 2018 ; P-1, 4/60, –2]**

(A) If $I_1 = I_2$, then $\vec{B}$ **cannot** be equal to zero at the origin (0, 0, 0)

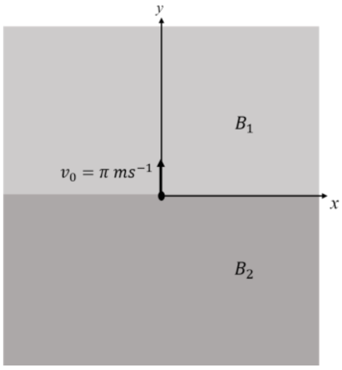
(B) If $I_1 > 0$ and $I_2 < 0$, then $\vec{B}$ can be equal to zero at the origin (0, 0, 0)

(C) If $I_1 < 0$ and $I_2 > 0$, then $\vec{B}$ can be equal to zero at the origin (0, 0, 0)

(D) If $I_1 = I_2$, then the z-component of the magnetic field at the centre of the loop is $\left(-\frac{\mu_0 I}{2R}\right)$

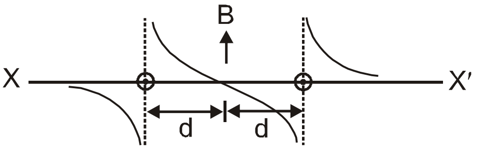
22. In the xy-plane, the region y > 0 has a uniform magnetic field $B_1\hat{k}$ and the region y < 0 has another uniform magnetic field $B_2\hat{k}$. A positively charged particle is projected from the origin along the positive y-axis with speed $v_0 = \pi ms^{-1}$ at t = 0, as shown in the figure. Neglect gravity in this problem. Let t = T be the time when the particle crosses the x-axis from below for the first time. If $B_2 = 4B_1$, the average speed of the particle, in ms^{-1}, along the x-axis in the time interval T is __________.

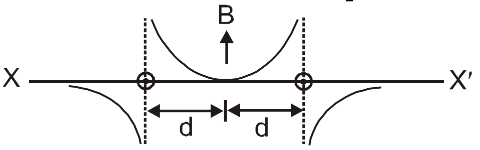
[JEE (Advanced) 2018 ; P-1, 3/60]

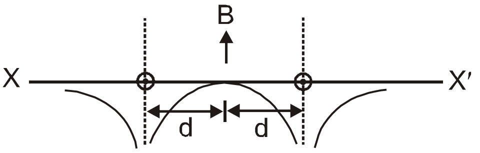


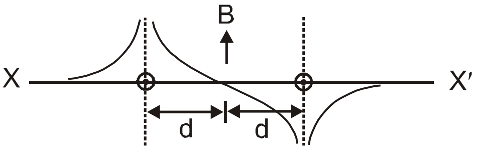
PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of magnetic field B along the line XX´ is given by **[AIEEE 4/144 2010]**

(1) 

(2) 

(3) 

(4) 

2.✍ A current I flows in an infinitely long wire with cross–section in the form of a semicircular ring of radius R. The magnitude of the magnetic induction on its axis is : **[AIEEE - 2011, 1-May, 4/120, –1]**

(1) $\frac{\mu_0 I}{\pi^2 R}$ (2) $\frac{\mu_0 I}{2\pi^2 R}$ (3) $\frac{\mu_0 I}{2\pi R}$ (4) $\frac{\mu_0 I}{4\pi R}$

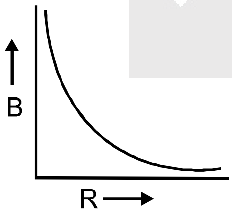
3. An electric charge +q moves with velocity $\vec{v} = 3\hat{i} + 4\hat{j} + \hat{k}$, in an electromagnetic field given by : $\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} + 3\hat{k}$. The y - component of the force experienced by + q is : **[AIEEE 2011, 11 May; 4/120, –1]**

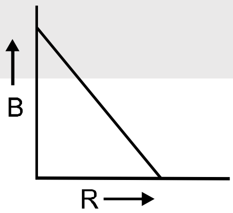
(1) 7 q (2) 5 q (3) 3 q (4) 2 q

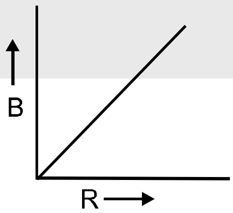
4. A thin circular disk of radius R is uniformly charged with density $\sigma > 0$ per unit area. The disk rotates about its axis with a uniform angular speed ω. The magnetic moment of the disk is : **[AIEEE 2011, 11 May; 4/120, –1]**

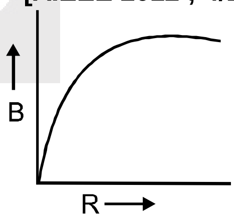
(1) $\pi R^4 \sigma\omega$ (2) $\frac{\pi R^4}{2}\sigma\omega$ (3) $\frac{\pi R^4}{4}\sigma\omega$ (4) $2\pi R^4 \sigma\omega$

5. A charge Q is uniformly distributed over the surface of non-condcting disc of radius R. The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω. As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. if we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure : **[AIEEE 2012 ; 4/120, –1]**

(1) 

(2) 

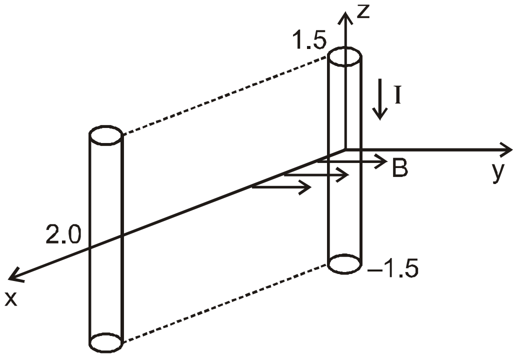
(3) 

(4) 

6. Two short bar magnets of length 1 cm each have magnetic moments 1.20 Am^2 and 1.00 Am^2 respectively. They are placed on a horizontal table parallel to each other with their N poles poining towards the South. They have a common magnetic equator and are separated by a distance of 20.0 cm. The value of the resultand horizontal magnetic induction at the mid - point O of the line joining their centres is close to (Horizontal component of earth's magnetic induction is 3.6×10^{-5} Wb/m^2) **[JEE (Main) 2013 ; 4/120, –1]**

(1) 3.6×10^{-5} Wb/m^2 (2) 2.56×10^{-4} Wb/m^2 (3) 3.50×10^{-4} Wb/m^2 (4) 5.80×10^{-4} Wb/m^2

7. A conductor lies along the z-axis at $-1.5 \le z < 1.5$ m and carries a fixed current of 10.0 A in $-\hat{a}_z$ direction (see figure). For a field $\vec{B} = 3.0 \times 10^{-4} e^{-0.2x}\, \hat{a}_y$ T, find the power required to move the conductor at constant speed to x = 2.0 m, y = 0 m in 5×10^{-3}s. Assume parallel motion along the x-axis **[JEE (Main) 2014, 4/120, –1]**



(1) 1.57 W (2) 2.97 W (3) 14.85 W (4) 29.7 W

8. Two coaxial solenoids of different radii carry current I in the same direction. Let $\vec{F}_1$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F}_2$ be the magnetic force on the outer solenoid due to the inner one. Then : **[JEE (Main) 2015; 4/120, –1]**

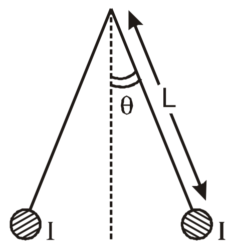
(1) $\vec{F}_1 = \vec{F}_2 = 0$

(2) $\vec{F}_1$ is radially inwards and $\vec{F}_2$ is radially outwards

(3) $\vec{F}_1$ is radially inwards and $\vec{F}_2 = 0$

(4) $\vec{F}_1$ is radially outwards and $\vec{F}_2 = 0$

9. Two long current carrying thin wires, both with current I, are held by insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle 'θ' with the vertical. If wires have mass λ per unit length then the value of I is : (g = gravitational acceleration) **[JEE (Main) 2015; 4/120, –1]**

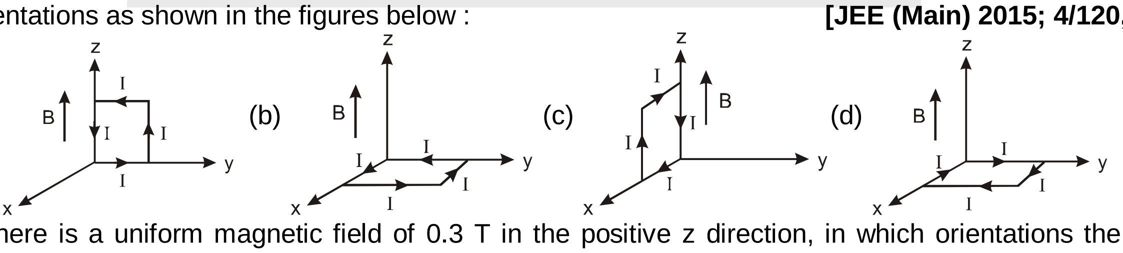


(1) $\sin\theta\sqrt{\dfrac{\pi\lambda gL}{\mu_0 \cos\theta}}$ (2) $2\sin\theta\sqrt{\dfrac{\pi\lambda gL}{\mu_0 \cos\theta}}$

(3) $2\sqrt{\dfrac{\pi gL}{\mu_0}\tan\theta}$ (4) $\sqrt{\dfrac{\pi\lambda gL}{\mu_0}\tan\theta}$

10. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is placed in different orientations as shown in the figures below : **[JEE (Main) 2015; 4/120, –1]**

(a) (b) (c) (d)



If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium ?

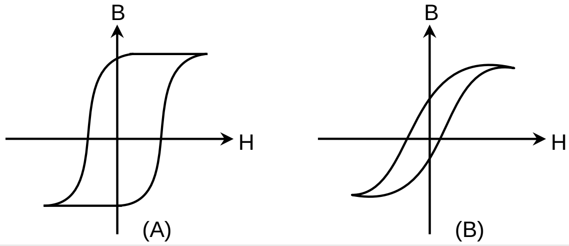
(1) (a) and (b), respectively (2) (a) and (c), respectively

(3) (b) and (d), respectively (4) (b) and (c), respectively

11. Two identical wires A and B, each of length l, carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side 'a'. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is : **[JEE (Main) 2016; 4/120, –1]**

(1) $\frac{\pi^2}{16\sqrt{2}}$ (2) $\frac{\pi^2}{16}$ (3) $\frac{\pi^2}{8\sqrt{2}}$ (4) $\frac{\pi^2}{8}$

12. Hysteresis loops for two magnetic materials A and B are given below :



These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use : **[JEE (Main) 2016; 4/120, –1]**
(1) A for electromgnets and B for electric generators.
(2) A for transformers and B for electric generators.
(3) B for electromgnets and transformers.
(4) A for electric generators and transformers.

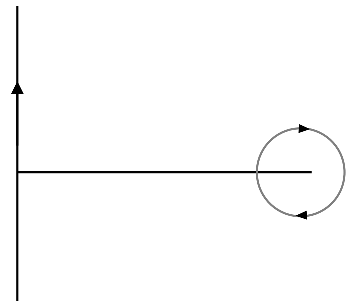
13. A magnetic needle of magnetic moment 6.7×10^{-2} Am^2 and moment of inertia 7.5×10^{-6} kg m^2 is performing simple harmonic oscillations in a magnetic field of 0.01 T. Time taken for 10 complete oscillations is : **[JEE (Main) 2017 ; 4/120, –1]**
(1) 8.76 s (2) 6.65 s (3) 8.89 s (4) 6.98 s

14. An electron, a proton and an alpha particle having the same kinetic energy are moving in circular orbits of radii r_e, r_p, r_α respectively in uniform magnetic field B. The relation between r_e, r_p, r_α is : **[JEE (Main) 2018; 4/120, –1]**
(1) $r_e < r_p < r_\alpha$ (2) $r_e < r_\alpha < r_p$ (3) $r_e > r_p = r_\alpha$ (4) $r_e < r_p = r_\alpha$

15. The dipole moment of a circular loop carrying a current I, is m and the magnetic field at the centre of the loop is B_1. When the dipole moment is doubled by keeping the current constant, the magnetic field at the centre of loop is B_2. The ratio $\frac{B_1}{B_2}$ is : **[JEE (Main) 2018; 4/120, –1]**

(1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$ (3) 2 (4) $\sqrt{3}$

16. An infinitely long, current carrying wire and a small current carrying loop are in the plane of the paper as shown. The radius of the loop is a and distance of its centre from the wire is d($d \gg a$). If the loop applies a force F on the wire then : **[JEE (Main) 2019; 4/120, –1]**



(1) F = 0
(2) $F \propto \left(\frac{a}{d}\right)^2$
(3) $F \propto \left(\frac{a^2}{d^3}\right)$
(4) $F \propto \left(\frac{a}{d}\right)$

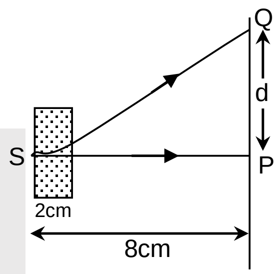
17. An insulating thin rod of length ℓ has a linear charge density $\rho(x) = \rho_0 \frac{x}{\ell}$ on it. The rod is rotated about an axis passing through the origin (x = 0) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is : **[JEE (Main) 2019; 4/120, –1]**

(1) $\frac{\pi}{4}n\rho\ell^3$ (2) $\pi n\rho\ell^3$ (3) $\frac{\pi}{3}n\rho\ell^3$ (4) $n\rho\ell^3$

18. A paramagnetic substance in the form of a cube with sides 1 cm has a magnetic dipole moment of 20×10^{-6}J/T when a magnetic intensity of 60×10^3 A/m is applied. Its magnetic susceptibility is **[JEE (Main) 2019; 4/120, –1]**

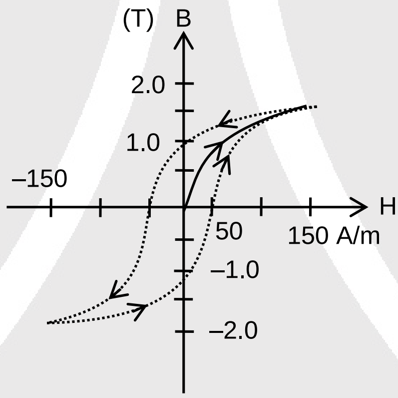
(1) 4.3×10^{-2} (2) 2.3×10^{-2} (3) 3.3×10^{-4} (4) 3.3×10^{-2}

19. An electron, moving along the x-axis with an initial energy of 100 eV, enters a region of magnetic field $\vec{B} = (1.5\times10^{-3}\,T)\hat{k}$ at S (See figure). The field extends between x = 0 and x = 2cm. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance d between P and Q (on the screen) is: (electron's charge = 1.6×10^{-19}C, mass of electron = 9.1×10^{-31} kg) **[JEE (Main) 2019; 4/120, –1]**



(1) 12.87 cm (2) 1.22 cm
(3) 2.25 cm (4) 11.65 cm

20. The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, co-ercivity and saturation, respectively, of the material are : **[JEE (Main) 2020, 07 January; 4/100, –1]**



(1) 1.0 T, 50 A/m and 1.5 T (2) 150 A/m, 1.0 T and 1.5 T
(3) 1.5 T, 50 A/m and 1.0 T (4) 1.5T, 50 A/m and 1.0 T

21. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0\hat{j}$. If an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{i}$ act on the particle, its speed will double after a time : **[JEE (Main) 2020, 07 January; 4/100, –1]**

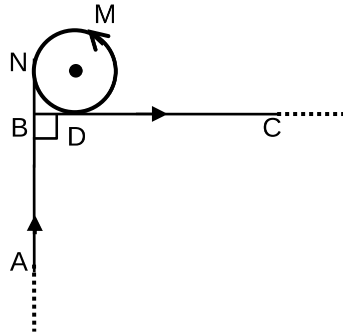
(1) $\frac{2mv_0}{qE_0}$ (2) $\frac{3mv_0}{qE_0}$ (3) $\frac{\sqrt{2}mv_0}{qE_0}$ (4) $\frac{\sqrt{3}mv_0}{qE_0}$

22. Proton with kinetic energy of 1 MeV moves from south to north. It gets an acceleration of 10^{12}m/s^2 by an applied magnetic field (west to east). The value of magnetic field: (Rest mass of proton is 1.6×10^{-27} kg) **[JEE (Main) 2020, 08 January; 4/100, –1]**

(1) 7.1 mT (2) 71 mT (3) 0.71 mT (4) 0.071 mT

23. A very long wire ABDMNDC is shown in figure carrying current I. AB and BC parts are straight, long and at right angle. At D wire forms a circular turn DMND of radius R. AB, BC parts are tangential to circular turn at N and D. Magnetic field at the centre of circle is :

[JEE (Main)2020, 08 January; 4/100, –1]



(1) $\frac{\mu_0 I}{2\pi R}\left(\pi+\frac{1}{\sqrt{2}}\right)$

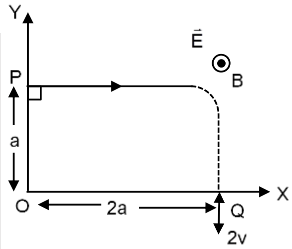
(2) $\frac{\mu_0 I}{2R}$

(3) $\frac{\mu_0 I}{2\pi R}\left(\pi-\frac{1}{\sqrt{2}}\right)$

(4) $\frac{\mu_0 I}{2\pi R}(\pi+1)$

24. A charged particle of mass 'm' and charge 'q' moving under the influence of uniform electric field $E\vec{i}$ and a uniform magnetic field $B\vec{k}$ follows a trajectory from point P to Q as shown in figure. The velocities at P and Q are respectively, $v\vec{i}$ and $-2v\vec{j}$. Then which of the following statements (A, B, C, D) are the correct ? (Trajectory shown is schematic and not to scale)

[JEE (Main) 2020, 09 January; 4/100, –1]



(A) $E=\frac{3}{4}\left(\frac{mv^2}{qa}\right)$

(B) Rate of work done by the electric field at P is $\frac{3}{4}\left(\frac{mv^3}{a}\right)$

(C) Rate of work done by both the fields at Q is zero

(D) The difference between the magnitude of angular momentum of the particle at P and Q is 2mav.

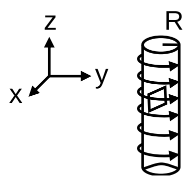
(1) (A), (B), (C) (2) (A), (B), (C), (D) (3) (A), (C), (D) (4) (B), (C), (D)

25. A long, straight wire of radius a carries a current distributed uniformly over its cross–section. The ratio of the magnetic fields due to the wire at distance $\frac{a}{3}$ and 2a, respectively from the axis of the wire is :

[JEE (Main) 2020, 09 January; 4/100, –1]

(1) $\frac{1}{2}$ (2) 2 (3) $\frac{3}{2}$ (4) $\frac{2}{3}$

26. An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has n turns/length and carries a current I. The electron gun shoots an electron along the radius of the solenoid with speed v. If the electron does not hit the surface of the solenoid, maximum possible value of v is (all symbols have their standard meaning) **[JEE (Main) 2020, 09 January; 4/100, –1]**



(1) $\frac{e\mu_0 nIR}{2m}$ (2) $\frac{e\mu_0 nIR}{m}$ (3) $\frac{2e\mu_0 nIR}{m}$ (4) $\frac{e\mu_0 nIR}{4m}$

27. A small circular loop of conducting wire has radius a and carries current I. It is placed in a uniform magnetic field B perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T. If the mass of the loop is m then :

[JEE Main) 2020, 09 January; 4/100, –1]

(1) $T=\sqrt{\frac{\pi m}{IB}}$ (2) $T=\sqrt{\frac{2m}{IB}}$ (3) $T=\sqrt{\frac{\pi m}{2IB}}$ (4) $T=\sqrt{\frac{2\pi m}{IB}}$

Answers

EXERCISE-1

PART-I

Section (A) :

A-1. $\frac{\sqrt{13}}{2} \times 10^{-4}$ wb/m^2

A-2. (i) 0, (ii) $10^{-13}\hat{k}$ (iii) $-10^{-13}\hat{j}$ (iv) $\frac{4}{27} \times 10^{-13}$ $(-2\hat{j} + \hat{k})$ (v) yes, no (vi) yes, no.

A-3. (i) $\frac{\mu_0 qv}{4\pi R^2}$, inwards (ii) $\frac{\mu_0 qv}{4\pi(x^2+R^2)}$, No

Section (B) :

B-1. 1×10^{-4} wb/m^2, towards the reader

B-2. 4×10^{-5} wb/m^2

B-3. (i) $2\sqrt{3} \times 10^{-5}$ T (ii) 2×10^{-5} T

B-4. 0

B-5. $\frac{\mu_0 i}{4\pi d}$, $\sqrt{2}\,\frac{\mu_0 i}{4\pi d}$, $\frac{1}{\sqrt{2}}$

B-6. $\frac{\mu_0 i}{2\pi x}$

B-7. $\frac{2\sqrt{2}\mu_0 i}{\pi a}$

B-8. 0

Section (C) :

C-1. (i) (a) $4\pi \times 10^{-4}$ T (b) zero

(ii) $2\sqrt{2}\,\pi \times 10^{-4}$ T

C-2. **(a)** $5\pi \times 10^{-5}\left(1-\frac{8}{13\sqrt{13}}\right)$T **(b)** zero

Section (D) :

D-1. $\frac{\mu_0 I}{4}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

D-2. $B = \frac{\mu_0 i}{4\pi R}\sqrt{2\left(2\pi^2+2\pi+1\right)}$

D-3. (a) $B = \frac{\mu_0 I}{4\pi}\left(\frac{2\pi-\varphi}{a}+\frac{\varphi}{b}\right)$;

(b) $B = \frac{\mu_0 I}{4\pi}\left(\frac{3\pi}{2a}+\frac{\sqrt{2}}{b}\right)$ (c)

$B = (\pi - \varphi + \tan\varphi)\,\mu_0 I/2\pi R = 28\mu T$.

D-4. (a) $\vec{B} = \frac{\mu_0 I}{4\pi R}(-\pi\hat{i} - 2\hat{k})$

(b) $\vec{B} = \frac{\mu_0 I}{4R}\left[-\hat{i}\ \left(1+\frac{1}{\pi}\right)-\frac{\hat{k}}{\pi}\right]$

(c) $\vec{B} = \frac{\mu_0 I}{4\pi R}\left(-\hat{j}-\hat{k}\right)$

Section (E) :

E-1. $B = \frac{\mu_0 ni}{2}$

E-2. $B = \frac{\mu_0 i}{2\pi b^2}r$

E-3. (a) zero (b) $\frac{2\mu_0 i}{5\pi r}$

E-4. 2500 turns/m

E-5. 1 V

E-6 (a) $\frac{K}{2n}$ (b) $\frac{\mu_0 K}{2}\sqrt{3}$, $\frac{\mu_0 K}{2}$

E-7. 0, μ_0 K towards right in the figure, 0

Section (F) :

F-1. 12 cm

F-2. D – electron , B – α-particle

F-3. $M^{-3} L^{-2} T^8 A^4$

F-4. **(a)** 4 A

(b) **(i)** current directed into the plane of paper 1 m from R on RQ (away from Q)

(ii) current directed out of from paper, 1 m from R on RQ (between R and Q)

F-5. $(-75\hat{i} + 100\hat{j})$ m/s

F-6. 3.0

F 7. (a) $\frac{mv}{qB}$ (b) $\pi/2$ (c) $\frac{\pi m}{2qB}$ (d) $\frac{mv}{qB}, \frac{3\pi}{2}, \frac{3\pi m}{2qB}$

F 8. (a) $\pi/2$ (b) $\pi/4$ (c) π

F-9. 8 cm

F-10. 50 m/s

F-11. $\frac{\mu_0 qrni}{2m}$

Section (G) :

G-1. 36 cm, $4\pi\sqrt{19}$ cm

G-2. $\frac{4}{\pi} \times 10^5$ m/s, 2×10^5 m/s

G-3. $15\pi \times 10^{-4}$ T

Section (H) :

H-1. 16×10^6 m/s, $\frac{91}{20}$ cm

H-2. (a) $\frac{i}{\pi r^2 ne}$ (b) $\frac{iB}{\pi r^2 n}$ upwards in the figure (c) $\frac{iB}{\pi r^2 ne}$ (d) $\frac{2iB}{\pi rne}$

H-3. $\frac{5}{4} \times 10^5$ C/kg

H-4. 5×10^3 N/C. 5×10^{-2} T

H-5. $\sqrt{\frac{2qE_0x}{m}}$

Section (I) :

I-1. 6×10^{-2} N perpendicular to both the wire and the field

I-2. 1×10^{-2} N on each wire, on da and cb towards left and on dc and ab downward.

I-3. 8×10^{-2} N

I-4. $\sqrt{2}B_0 i\ell$

I-5. 5×10^{-4} T, horizontal and $\perp$ to the wire.

I-6. $i\lambda B/2$

I-7. $-iRB\hat{j}$,

I-8. 0.12

I-9. The current carrying wires are electrically neutral. Hence the only interaction between wires is of attractive magnetic force.
But in parallel beams of electrons both the beams have negative charge.
Hence there is electrostatic repulsion and also magnetic attraction. Electrostatic repulsion has larger magnitude than magnetic attraction. Hence the beams repel.

I-10. $iB_0 \ell$

I-11. 6×10^{-3} N/m, downward zero

I-12. $\frac{\mu_0 i_1 i_2}{2\pi} \ell n \frac{r_2}{r_1} = 40\ \ell n2\ \mu$ J/m

I-13. $\frac{\mu mg}{i\ell\sqrt{1+\mu^2}}$

Section (J) :

J-1. $\pm 75\pi \times 10^{-3}$ J

J-2. (a) $2\pi aiB$, perpendicular to the plane of the figure going into it. (b) $\sqrt{2}\pi ai \cdot B_0$

J-3 $\frac{1}{2}$T

J-4 (a) $4\pi \times 10^{-2}$ N-m (b) 60^o

J-5 (a) zero (b) 2×10^{-2} N-m parallel to the side.

J-6. $\pi \times 10^{-2}$ N - m

J-7. (a) $\frac{iL^2B}{4\pi}$ (b) $\frac{BiL^2}{18}$

J-8. $\frac{1}{2}Q\omega R^2$

Section (K) :

K-1. $I_V = 1112 \times 10^{-4}$ A, $I_H = 556\sqrt{3} \times 10^{-4}$ A.

K-2. $3\sqrt{31} \times 10^{-5}$ T, $\tan^{-1}\left(\frac{5\sqrt{3}}{7}\right)$, East of north

K-3. (a) 1.0 A, (b) 2.0 V perpendicular to the magnetic meridian and in vertical plane

Section (L) :

L-1. 7.5×10^{-4} N/A^2, 596 **L-2.** $\mu_r = 798.6$

PART-II

Section (A) :

A-1.	(B)	**A-2.**	(A)

Section (B) :

B-1.	(A)	**B-2.**	(B)	**B-3.**	(C)
B-4.	(C)	**B-5.**	(B)	**B-6.**	(C)

Section (C) :

C-1.	(A)	**C-2.**	(B)	**C-3.**	(D)
C-4.	(D)	**C-5.**	(A)	**C-6.**	(B)
C-7.	(B)	**C-8.**	(C)	**C-9.**	(D)

Section (D) :

D-1.	(B)	**D-2.**	(D)	**D-3.**	(C)
D-4.	(B)	**D-5.**	(D)	**D-6.**	(A)

Section (E) :

E- 1. (C)

Section (F) :

F-1.	(D)	**F-2.**	(B)

Section (G) :

G-1.	(C)	**G-2.**	(C)	**G-3.**	(B)
G-4.	(B)	**G-5.**	(D)	**G-6.**	(C)
G-7.	(C)	**G-8.**	(B)	**G-9.**	(A)

Section (H) :

H-1.	(B)	**H-2.**	(A)	**H-3.**	(B)

Section (I) :

I-1.	(C)	**I-2.**	(A)

Section (J) :

J-1	(C)	**J-2**	(C)	**J-3**	(B)
J-4	(B)	**J-5.**	(C)	**J-6**	(C)
J-7	(A)	**J-8**	(A)	**J-9**	(B)
J-10	(B)				

PART-III

1. (A) p,q, r, t (B) p, q, r, s, t (C) r (D) p, q, r, s, t
2. (A) p, q, t (B) p, r (C) p (D) p, q, s
3. (A) r, (B) q,u (C) u, (D) t

EXERCISE-2

PART-I

1.	(B)	**2.**	(A)	**3.**	(C)
4.	(C)	**5.**	(A)	**6.**	(C)
7.	(A)	**8.**	(B)	**9.**	(B)
10.	(A)	**11.**	(A)	**12.**	(B)
13.	(D)	**14.**	(B)	**15.**	(D)
16.	(A)	**17.**	(C)	**18.**	(B)
19.	(B)	**20.**	(B)		

PART-II

1.	6	**2.**	N = 2	**3.**	1
4.	3	**5.**	16	**6.**	32
7.	10	**8.**	25	**9.**	2
10.	2	**11.**	5	**12.**	4
13.	2	**14.**	4		
15.	(a) 2		(b) 8	**16.**	4
17.	8	**18.**	(a) 3	(b) 4	
19.	(a) 7	(b) 15			
20.	(a) 20		(b) 2	**21.**	5

PART-III

1.	(A,B,C)	**2.**	(C,D)	**3.**	(A,D)
4.	(B,C,D)	**5.**	(B,C)	**6.**	(A,B,D)
7.	(A,B,C)	**8.**	(B,D)	**9.**	(A,C)
10.	(C,D)	**11.**	(B,D)	**12.**	(C,D)
13.	(A,C,D)	**14.**	(B,D)	**15.**	(A,C)
16.	(B,C)	**17.**	(A,B)	**18.**	(A,D)

PART-IV

1.	(B)	**2.**	(A)	**3.**	(A)
4.	(A)	**5.**	(C)	**6.**	(C)
7.	(A)	**8.**	(A)	**9.**	(D)
10.	(C)	**11.**	(B)	**12.**	(A)
13.	(D)				

EXERCISE-3

PART-I

1.	(C)	**2.**	(B,D)	**3.**	(A)
4.	(C,D)	**5.**	5	**6.**	(B)
7.	(D)	**8.**	(A,C)	**9.**	(A,C)
10.	3	**11.**	(C)	**12.**	(B)
13.	(A,B,C)	**14.**	(A,D)	**15.**	(A,C)
16.	(C)	**17.**	(A)	**18.**	(A)
19.	(A)	**20.**	(C,D)	**21.**	(A,B,D)
22.	2.00				

PART-II

1.	(1)	**2.**	(1)	**3.**	(1)
4.	(3)	**5.**	(1)	**6.**	(2)
7.	(2)	**8.**	(4)	**9.**	(2)
10.	(3)	**11.**	(3)	**12.**	(3)
13.	(2)	**14.**	(4)	**15.**	(1)
16.	(2)	**17.**	(1)	**18.**	(3)
19.	(1)	**20.**	(1)	**21.**	(4)
22.	(3)	**23.**	(1)	**24.**	(1)
25.	(4)	**26.**	(1)	**27.**	(4)